

Technical Assessment

Cloud & Network Security

Cheatsheet for Accenture

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1. BASICS OF NETWORKING

1.1 Introduction to Networking

Computer Networking is the process of connecting two or more computing devices so they can **communicate and share data, resources, and services** with each other.

Why do we use Computer Networks?

-  **File & Data Sharing** – Share files and information between devices.
-  **Resource Sharing** – Share printers, storage, applications, etc.
-  **Communication** – Enables email, messaging, video conferencing, etc.
-  **Internet Access** – Allows multiple devices to access network/internet services.
-  **Centralized Management** – Resources and security can be managed centrally.
-  **Cost Efficiency** – Shared resources can reduce infrastructure costs.

◆ Network Components

A computer network consists of several basic components that work together to enable communication.

Component	What it does
Sender	Device that sends data
Receiver	Device that receives data
Transmission Medium	Path through which data travels
Network Devices	Connect/manage network communication
Protocols	Rules governing communication
Network Interface	Enables a device to connect to a network

Examples

Transmission Medium

- Wired → Ethernet cable, Fiber-optic cable
- Wireless → Wi-Fi, Radio waves

Network Devices

- Switch, Router, Hub, Modem, Access Point, Firewall

Protocols

- TCP/IP
- HTTP/HTTPS
- FTP
- DNS

Nodes, Hosts & Clients

Quick Revision:

Term	Meaning	Example
Node	Any participating network point/device	Router, PC
Host	Network-connected device capable of communication	PC, Server
Client	Requests service/resource	Browser, Mobile App
Server	Provides service/resource	Web Server, Mail Server

Wired vs Wireless Networking

Networks can transmit data using either **physical cables** or **wireless signals**.

Feature	Wired Networking	Wireless Networking
Medium	Physical cable	Radio/wireless signals
Examples	Ethernet, Fiber	Wi-Fi, Bluetooth
Speed	Generally high & stable	Can vary with conditions
Mobility	Limited	High
Interference	Generally lower	More susceptible
Installation	Requires cabling	Easier to deploy

Security	Physical access adds protection	Requires strong wireless security
Common Use	Offices, data centers	Homes, phones, public networks

Wired Networking

Uses a physical transmission medium.

Examples:

- Ethernet
- Fiber Optic

Wireless Networking

Uses electromagnetic/radio signals rather than a physical cable.

Examples:

- Wi-Fi
- Bluetooth
- Cellular networks

##Data Communication Basics

Data Communication is the exchange of data between two or more devices through a communication medium.

Basic Data Communication System:-

Sender → Transmission Medium → Receiver

Example:

Laptop → Wi-Fi → Router → Internet Server

Element	Meaning
Sender	Generates/sends the data
Receiver	Receives the data

Message	Information being transmitted
Transmission Medium	Path used to transmit data
Protocol	Rules governing communication

1.2 Types of Networks

Networks are commonly classified based on **geographical coverage, size, ownership, and purpose**. The four traditional categories are **PAN, LAN, MAN, and WAN**.

1. PAN — Personal Area Network

A Personal Area Network (PAN) is a small network used to connect devices around an individual, typically within a short range.

Key Points

- Smallest commonly discussed network type.
- Usually covers a few meters around a person.
- Connects personal devices such as smartphones, laptops, smartwatches and earbuds.
- Can use **Bluetooth, USB, NFC**, etc.
- Generally designed for personal use.

Example

Connecting your **smartphone to Bluetooth earbuds or a smartwatch**.



Remember

PAN = Personal devices + Very short range

2. LAN — Local Area Network

A **Local Area Network (LAN)** connects devices within a **limited geographical area**, such as a home, office, laboratory, school or college campus.

Key Points

- Covers a relatively small geographical area.
- Usually managed by a single organization, institution or individual.

- Common technologies include **Ethernet and Wi-Fi**.
- Generally provides high-speed communication within the local network.

Example: Computers, printers and servers connected within a **college computer laboratory**.

Advantages

- High-speed local communication
- Resource sharing
- Relatively easy to manage
- Lower cost compared with large-area networks

Remember

LAN = Local area → Home / Office / Lab

3. MAN — Metropolitan Area Network

A **Metropolitan Area Network (MAN)** connects multiple LANs across a **city or metropolitan-scale area**.

It is larger than a LAN but generally smaller than a WAN.

Key Points

- Covers a city or metropolitan region.
- Can connect networks belonging to different buildings or locations.
- Often uses high-capacity links such as **fiber-optic connections**.
- Can be operated by organizations, institutions, service providers or government bodies.

Example

A university network connecting campuses located at different locations **across the same city**.

Remember

MAN = Metropolitan / City-level network

4. WAN — Wide Area Network

A **Wide Area Network (WAN)** connects networks across a **large geographical area**, potentially spanning cities, countries or continents.

Key Points

- Largest geographical coverage among PAN, LAN and MAN.
- Connects multiple LANs/MANs over long distances.
- May use leased lines, fiber-optic links, cellular networks, satellite links and other long-distance communication technologies.
- Often involves multiple network providers and routing infrastructure.

Example

A multinational company's network connecting offices in **India, the USA, Europe and other countries**.

 **The Internet** is the best-known example of a global interconnected network and is commonly used as an example of WAN-scale networking.

PAN vs LAN vs MAN vs WAN

Feature	PAN	LAN	MAN	WAN
Full Form	Personal Area Network	Local Area Network	Metropolitan Area Network	Wide Area Network
Coverage	Very small	Building / local site	City / metropolitan area	Large geographical area
Typical Example	Phone + earbuds	College lab	City-wide campus network	Multinational network
Common Technologies	Bluetooth, NFC, USB	Ethernet, Wi-Fi	Fiber / high-capacity links	Fiber, leased lines, cellular, satellite
Relative Size	★ Smallest	Small	Medium	★ Largest
Typical Use	Personal devices	Homes/offices	City-wide connectivity	Long-distance connectivity

1.3 Network Addressing

Network addressing is used to **identify devices and interfaces on a network** so that data can be delivered to the correct destination.

The two most important addressing concepts for this section are:

- **IP Address → Logical addressing**
 - **MAC Address → Hardware/link-layer addressing**
-

1. IP Address

An **IP (Internet Protocol) address** is a logical address assigned to a network interface so that devices can be **identified and reached across an IP network**.

Why is an IP Address needed?

When data travels from one network to another, routers need an address to determine **where the packet should go**.

Key Points

- Used for **logical addressing**.
- Two major versions:
 - **IPv4**
 - **IPv6**

Example

192.168.1.10 → IPv4 address

2001:db8::1 → IPv6 address

Remember

IP = Logical Address + Network Routing

2. IPv4

IPv4 (Internet Protocol version 4) is the fourth version of the Internet Protocol and remains widely deployed.

Key Characteristics

- Uses **32-bit addresses**.
- Written in **dotted-decimal notation**.
- Divided into **4 octets**.
- Each octet ranges from **0 to 255**.
- Example: 192.168.1.10

Example

192.168.1.25

Octet	Value
1st	192
2nd	168
3rd	1
4th	25

3. IPv6

IPv6 (Internet Protocol version 6) was developed to address the limitations of IPv4, especially its limited address space.

Key Characteristics

- Uses **128-bit addresses**.
- Written in **hexadecimal notation**.
- Address is represented as **8 groups of 16 bits**.
- Groups are separated using colons (:).
- Example:

2001:0db8:85a3:0000:0000:8a2e:0370:7334

4. IPv4 vs IPv6

Feature	IPv4	IPv6
Address Size	32-bit	128-bit

Notation	Dotted decimal	Hexadecimal
Example	192.168.1.1	2001:db8::1
Address Space	2^{32}	2^{128}
Header	Variable length (20–60 bytes)	Fixed 40 bytes
Broadcast	Supported	No traditional broadcast
Address Resolution	ARP	Neighbor Discovery (NDP)
Address Configuration	Manual / DHCP	Manual / SLAAC / DHCPv6
Fragmentation	Can be performed by sender and routers	Performed by source using fragmentation extension header
Checksum	Header checksum present	No IP header checksum
Security	IPsec support	IPsec support

5. MAC Address

A **MAC (Media Access Control) address** is a link-layer address associated with a network interface and is used for communication within a local network.

It operates at the **Data Link Layer (Layer 2)** of the OSI model.

Key Characteristics

- Used for **link/local network communication**.
- Typically **48 bits (6 bytes)** for standard Ethernet MAC addresses.

Important Clarification

A MAC address should **not be treated as an absolutely permanent identity**.

Although a manufacturer-assigned MAC address is typically associated with a network interface, MAC addresses can be **locally administered, changed/spoofed, or randomized**.

6. IP Address vs MAC Address

This is a very important MCQ comparison.

Feature	IP Address	MAC Address
Type	Logical address	Link-layer/hardware address
Main Layer	Network Layer (L3)	Data Link Layer (L2)
Purpose	Identifies/reaches interfaces across IP networks	Local/link-layer delivery
Used By	Routers	Switches
Associated With	Network interface/network configuration	Network interface
Common Size	IPv4: 32-bit; IPv6: 128-bit	Typically 48-bit
Notation	IPv4: decimal; IPv6: hexadecimal	Hexadecimal
Example	192.168.1.10	00:1A:2B:3C:4D:5E
Can Change?	Yes, commonly	Can also be changed/randomized
Data Unit	IP Packet	Ethernet Frame

1.4 Networking Protocols

What is a Network Protocol?

A **network protocol** is a set of rules that defines how devices communicate and exchange data over a network.

- Defines how data is **formatted, transmitted, received, and interpreted**.
- Different protocols are designed for different purposes.
- Protocols operate at different layers of networking models.

Protocol = Rules for communication

TCP — Transmission Control Protocol

- **Transport Layer** protocol that provides reliable communication between applications.
- It is **connection-oriented**, meaning a connection is established before data transfer.

- Ensures **ordered and reliable delivery** using acknowledgements and retransmission.
- Provides **flow control and congestion control**.
- Used where data loss or incorrect ordering is unacceptable.
- **Examples:** Web communication, file transfer, email.

TCP = Reliable + Ordered + Connection-oriented

UDP — User Datagram Protocol

- **Transport Layer** protocol that provides connectionless communication.
- Does not guarantee **delivery, ordering, or retransmission** of packets.
- Has lower overhead than TCP because it does not establish a connection or provide TCP-style reliability.
- Suitable for applications where **low latency** is more important than guaranteed delivery.
- Commonly used in **DNS, online gaming, VoIP and real-time communication**.
- Uses **datagrams** for data transmission.

UDP = Connectionless + Low Overhead + No Delivery Guarantee

TCP vs UDP — Quick View

HTTP — HyperText Transfer Protocol

- **Application Layer** protocol used for communication between **web clients and web servers**.
- Follows a **request-response** communication model.
- Used to transfer web resources such as HTML pages, images and other data.
- HTTP itself does **not encrypt** the communication.

HTTP = Web Communication

HTTPS — HyperText Transfer Protocol Secure

- HTTPS is **HTTP protected by TLS (Transport Layer Security)**.
- Provides **confidentiality and integrity** for data transmitted between client and server.
- TLS certificates can also help authenticate the server.
- Commonly used for **secure websites, online banking, payments and logins**.
- HTTPS does not guarantee that the website itself is trustworthy; it secures the connection.

HTTPS = HTTP + TLS Security

FTP — File Transfer Protocol

- **Application Layer** protocol designed for transferring files between a client and server.
- Supports operations such as **uploading, downloading and managing files**.
- Traditional FTP does **not encrypt** credentials or transferred data.
- For secure file transfer, **SFTP or FTPS** can be used.

FTP = File Transfer

SFTP — SSH File Transfer Protocol

- Provides **secure file transfer using SSH**.
- Encrypts authentication credentials and transferred data.
- Uses a **single SSH connection**, typically over **TCP port 22**.
- Supports secure file upload, download and file management.

- SFTP is **not simply a secure version of FTP**; it is a separate protocol built on SSH.
- Commonly used when files need to be transferred securely over an untrusted network.

SFTP = Secure File Transfer through SSH

SMTP — Simple Mail Transfer Protocol

- **Application Layer** protocol primarily used for **sending and relaying email**.
- Used when an email client submits a message to a mail server.
- Also used for communication between mail servers during email delivery.
- SMTP is mainly associated with **outgoing mail**, not retrieving messages.
- Email retrieval is generally handled using **POP3 or IMAP**.

SMTP = Send Email

POP3 — Post Office Protocol Version 3

- **Application Layer** protocol used to retrieve email from a mail server.
- Primarily designed around **downloading messages** to an email client.
- Traditionally, downloaded messages may be removed from the server, although clients can be configured to keep copies.
- Less suitable for synchronizing the same mailbox across multiple devices.
- Common ports: **110** (standard) and **995** (POP3 over TLS).
- Best remembered as a **download-oriented email protocol**.

POP3 = Download Email

IMAP — Internet Message Access Protocol

- **Application Layer** protocol used to access and manage email stored on a mail server.
- Keeps messages primarily on the server and supports **synchronization across multiple devices**.
- Synchronizes folders, read/unread status and other mailbox information.
- Better suited than POP3 when using email on **phones, laptops and multiple devices**.
- Common ports: **143** (standard) and **993** (IMAP over TLS).
- Commonly used by modern email services.

IMAP = Synchronize & Manage Email

DNS — Domain Name System

- DNS is a **distributed naming system** that translates domain names into IP addresses and provides other DNS records.
- Example: `www.example.com` → corresponding IP address.
- Makes websites easier to access using **names instead of numerical IP addresses**.
- DNS is often described as the “**phonebook of the Internet**.”

DNS = Domain Name → IP Address

Telnet

- **Application Layer** protocol used for **remote terminal access** to another system.
- Allows a user to interact with a remote machine through a command-line interface.
- Uses **TCP port 23**.

- Traditional Telnet sends communication, including credentials, **without encryption**.
- Therefore, it is considered **insecure for remote administration**.
- **SSH** is the preferred secure alternative.

Telnet = Remote Access, but NOT Secure

Protocols: Must Remember

TCP → Reliable communication

UDP → Connectionless communication

HTTP → Web

HTTPS → Secure Web

FTP → File Transfer

SFTP → Secure File Transfer

SMTP → Send Email

POP3 → Download Email

IMAP → Synchronize Email

DNS → Domain Name Resolution

Telnet → Remote Access

Common Terms related to data :

Bandwidth

Bandwidth is the **maximum data-carrying capacity** of a network connection or communication channel.

- Usually measured in **bits per second (bps)**.
- Common units: **Mbps, Gbps**.
- Higher bandwidth means the connection can potentially carry **more data per unit of time**.
- Bandwidth represents **capacity**, not necessarily the actual data transfer rate.
- A high-bandwidth connection can still experience low performance because of latency, congestion or other factors.

Latency

Latency is the **time taken for data to travel from one point to another.**

- Usually measured in **milliseconds (ms).**
- Lower latency means faster response.
- Important for applications requiring quick responses, such as **online gaming, video calls and remote systems.**
- Latency can be affected by distance, network congestion, processing and transmission conditions.
- **Ping time** is commonly used as a practical measure of network round-trip latency.

Latency = Delay

Throughput

Throughput is the **actual rate at which data is successfully transferred over a network.**

- Usually measured in **bps, Mbps or Gbps.**
- Unlike bandwidth, throughput represents the **actual achieved data transfer rate.**
- It can be lower than the available bandwidth because of congestion, protocol overhead, packet loss and network conditions.
- Higher throughput generally means more data is being successfully transferred per unit of time.

Bandwidth vs Throughput

Bandwidth = Maximum possible capacity

Throughput = Actual achieved transfer rate

Example:

A connection may have **100 Mbps bandwidth**, but due to congestion and overhead, its actual throughput may be **75 Mbps.**

7. Jitter

Jitter is the **variation in packet arrival time** during network communication.

Imagine packets being sent at regular intervals:

Packet 1 → Packet 2 → Packet 3 → Packet 4

If they arrive with inconsistent delays:

10 ms → 15 ms → 40 ms → 12 ms

the variation is called **jitter**.

- Usually measured in **milliseconds (ms)**.
- High jitter can cause **audio/video distortion, delays and interruptions**.
- Particularly important for **VoIP, video conferencing and real-time applications**.
- Lower and more consistent delay is desirable for real-time communication.

Jitter = Variation in delay

1.7 Network Communication Models

Network communication models describe **how devices communicate and share resources** with each other.

1. Client-Server Model

In the **Client-Server model**, clients request services or resources, while a **centralized server** provides those services.

Key Points

- **Client** → Requests a service/resource.
- **Server** → Processes the request and provides the service/resource.

- Server usually has a **centralized role** in managing data, applications, authentication, or resources.
- Multiple clients can communicate with the same server simultaneously.
- Common in websites, email systems, databases, cloud applications and enterprise networks.
- Requires proper server availability and management.

Example

When you open a website:

Browser (Client) → Request → Web Server

Browser (Client) ← Response ← Web Server

Advantages

- Centralized management
- Easier data/security control
- Centralized backup and resource management
- Suitable for large networks

Client requests → Server provides

🔗 2. Peer-to-Peer (P2P) Model

In a **Peer-to-Peer (P2P) network**, connected devices (peers) can **communicate directly and share resources with one another** without requiring a dedicated central server for every interaction.

Key Points

- Each peer can act as both a **client and a server**.
- Resources can be shared directly between peers.
- Does not require a dedicated central server for basic resource sharing.
- Generally suitable for **small networks or decentralized applications**.

- Management and security can become more difficult as the number of peers increases.
- **BitTorrent** is a well-known example of P2P file distribution.

Example

Computer A ↔ Computer B ↔ Computer C

Each computer can provide and request resources from other peers.

Peer ↔ Peer → Direct Resource Sharing

1.8 Switching, Routing & Data Delivery

These concepts describe **how data moves through a network**, from the source device toward its destination.

1. Switching

Switching is the process of **forwarding data from one device/interface to another within a network**.

Example

PC A → Switch → PC B

The switch examines the destination MAC address and forwards the Ethernet frame through the appropriate port.

Switching = Local forwarding using MAC addresses

2. Routing

Routing is the process of determining and selecting a **path for packets to travel between different networks**.

- Routing primarily operates at the **Network Layer (Layer 3)**.
- **Routers** use destination IP addresses to make forwarding decisions.
- A routing table contains information about available destinations and

Example

Network A → Router → Network B → Router → Network C

Each router examines the destination IP address and forwards the packet toward the appropriate next hop.

Routing = Path selection + IP addresses

2.1 Networking Devices

Networking devices are hardware or software components used to connect devices, forward data, control traffic, provide network access, or improve network security.

1. Modem

Modem stands for **Modulator-Demodulator**.

- Converts signals between the form used by the local network/device and the transmission technology of the ISP connection.
- Commonly connects a home/office network to an **ISP network**.
- A modem is **not primarily a routing device**, although many consumer devices combine modem + router functions.
- Its exact operation depends on the access technology, such as cable, DSL, or other broadband technologies.

2. Hub

A **Hub** is a basic networking device that connects multiple devices in a LAN and **broadcasts incoming data to all connected ports**.

- Operates at the **Physical Layer (Layer 1)**.
- Does not examine MAC addresses or determine the intended destination.
- When a signal arrives, it repeats it out through its other ports.
- All connected devices share the same collision domain in a traditional hub-based Ethernet network.
- Hubs are largely obsolete and have been replaced by switches.

3. Switch

A **Switch** connects devices within a LAN and intelligently **forwards Ethernet frames toward the appropriate destination port**.

- A traditional Ethernet switch primarily operates at the **Data Link Layer (Layer 2)**.
- Uses **MAC addresses** to make forwarding decisions.
- Learns MAC addresses by examining the **source MAC address** of incoming frames.
- Maintains a **MAC address/forwarding table**.
- Unlike a hub, it normally forwards a unicast frame only toward the port associated with the destination MAC.
- Each switch port typically forms a separate collision domain in modern switched Ethernet.

4. Router

A **Router** connects **different IP networks** and forwards packets toward their destination based primarily on **IP addresses**.

- Operates primarily at the **Network Layer (Layer 3)**.
- Uses a **routing table** to determine where packets should be forwarded.
- Each router interface generally belongs to a different IP network/subnet.
- Routers separate **broadcast domains**.
- Can connect a local network to other networks, including the Internet.
- Routers can use **static routes or dynamic routing protocols** to learn routes.

5. Bridge

A **Bridge** connects and filters traffic between **two or more LAN segments**.

- Traditionally operates at the **Data Link Layer (Layer 2)**.
- Uses **MAC addresses** to decide whether frames should be forwarded.
- Can reduce unnecessary traffic between network segments.
- Separates collision domains.
- Conceptually, a bridge performs a function similar to a basic multi-port switch.
- Modern switches largely replaced traditional standalone bridges.

6. Repeater

A **Repeater** receives a signal and **regenerates/reshapes it** so it can travel farther.

- Operates at the **Physical Layer (Layer 1)**.
- Does not inspect IP or MAC addresses.
- Does not make routing or forwarding decisions based on addresses.
- Used to overcome signal degradation over longer transmission distances.
- Works with the physical characteristics of the transmission medium.

7. Gateway

A **Gateway** is a device or system that acts as an **entry/exit point between networks or systems**, and in some contexts can perform protocol translation.

- The term **gateway is context-dependent** and is not limited to one OSI layer.
- A gateway can connect networks that use **different protocols or architectures** and may translate between them.

- In everyday IP networking, the **default gateway** is usually the router/interface through which a host reaches destinations outside its local subnet.
- Gateways can also perform application-level or protocol-level translation.
- Because of its broad meaning, simply saying “**Gateway = Layer 7**” is an oversimplification.

✓ **Gateway is a general term and can operate at different layers depending on its function.**



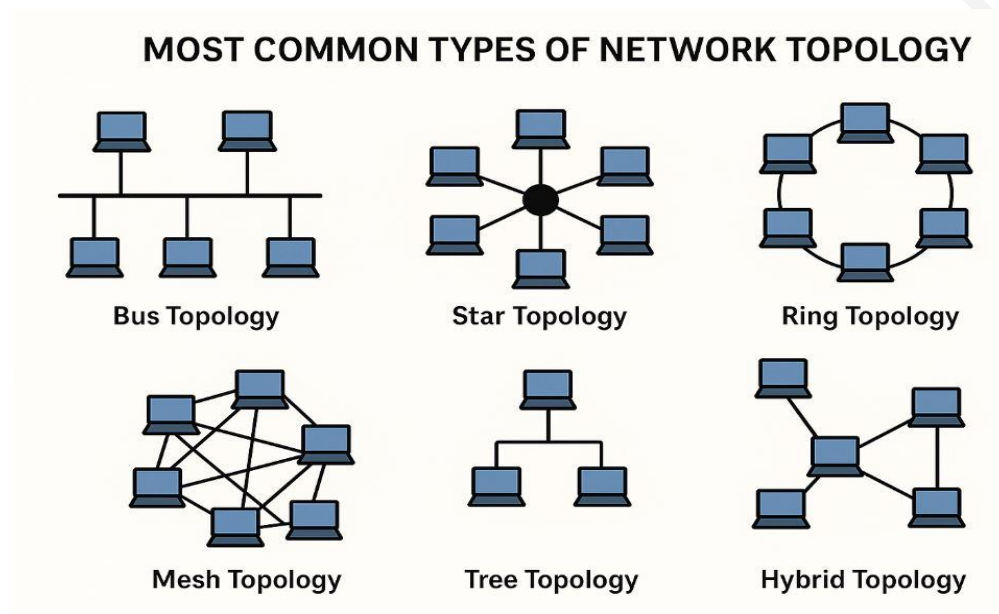
3. Network Topologies

3.1 Introduction

Network Topology refers to the physical or logical arrangement of devices (nodes) and connections within a computer network.

It describes how devices are connected and how data flows between them.

Topology = Arrangement of devices + Connections + Data flow



3.2 Types of Network Topologies

Network topology defines how devices and communication links are arranged. The major topologies are **Bus, Star, Ring, Mesh, Tree, and Hybrid**.

1. Bus Topology

In a **Bus Topology**, all devices are connected to a **single shared backbone cable**. Data transmitted onto the backbone is available to devices on the network, with the intended device accepting the relevant transmission.

Advantages

- Simple to install and understand.
- Requires **less cable** than many other topologies.
- Relatively low initial cost for small networks.

Limitations

- Failure of the **main backbone cable** can disrupt the entire network.
- Performance decreases as more devices and traffic are added.
- Troubleshooting can be difficult.
- Less scalable than modern switched networks.

Remember

Bus = One shared backbone

★ 2. Star Topology

In a **Star Topology**, every device is connected to a **central device**, usually a switch or hub.

Advantages

- Easy to install, manage and troubleshoot.
- Failure of one individual cable/device generally affects **only that device**.
- Easy to add or remove devices.
- Good performance with modern switches.

Limitations

- Failure of the **central device** can affect the entire network.
- Requires more cable than a bus topology.
- Central equipment adds cost.

Remember

Star = Central device

3. Ring Topology

In a **Ring Topology**, each device is connected to **two neighboring devices**, forming a closed loop.

The last device connects back to the first.

Advantages

- Data transmission can follow a predictable path.
- Can provide orderly access to the communication medium in some ring implementations.
- Each device has a defined connection to its neighbors.

Limitations

- Failure of a single link/device can disrupt communication in a simple single-ring design.
- Adding or removing devices can affect the network.
- Troubleshooting can be more difficult than in a star topology.



Remember

Ring = Closed loop

4. Mesh Topology

In a **Mesh Topology**, devices are connected through **multiple interconnections**, providing multiple possible paths for communication.

Advantages

- High **reliability and fault tolerance**.
- Multiple paths are available if one link fails.
- Suitable where network availability is critical.

Limitations

- Requires more cables/links and network interfaces.
- More expensive to implement and maintain.
- Full mesh becomes difficult to scale as the number of devices increases.



Remember

Mesh = Multiple paths + High redundancy

5. Tree Topology

A **Tree Topology** is a hierarchical topology that combines characteristics of **star and bus-style arrangements**.

Devices are organized into **levels or branches**, usually with higher-level devices connecting to lower-level network segments.

Advantages

- Supports **hierarchical network organization**.
- Easier to expand by adding branches/segments.
- Makes management of large structured networks easier.
- Faults can sometimes be isolated to a particular branch.

Limitations

- Failure of an important higher-level device/link can affect an entire branch.
- Requires more infrastructure than simple topologies.
- Configuration and management can become complex as the network grows.

Remember

Tree = Hierarchical + Branches

6. Hybrid Topology

A **Hybrid Topology** combines **two or more different topology types** within the same network.

For example:

Star + Bus = Hybrid Topology

A large organization might use star-based LANs in different departments and connect those segments using another network structure.

Advantages

- Highly **flexible and customizable**.

- Can be designed according to the requirements of different network segments.
- Suitable for large and complex networks.
- Easier to expand or modify individual sections.

Limitations

- More expensive than simple topologies.
- Design and configuration can be complex.
- Troubleshooting may require understanding multiple topology types.

Remember

Hybrid = Combination of topologies



Topology Quick Comparison

Topology	Key Structure	Major Advantage	Major Limitation
Bus	Single backbone	Low cable requirement	Backbone failure affects network
Star	Central device	Easy management & troubleshooting	Central device is critical
Ring	Closed loop	Predictable communication path	Link/device failure can disrupt simple ring
Mesh	Multiple interconnections	High redundancy & reliability	Expensive and complex
Tree	Hierarchical branches	Scalable hierarchical structure	Higher-level failure can affect branches
Hybrid	Combination	Flexible & customizable	Complex and costly

4.1 Networking Models

A **networking model** is a conceptual framework that divides network communication into **layers**, with each layer responsible for specific functions.

Why are networking models used?

- Break complex network communication into **smaller, manageable functions**.

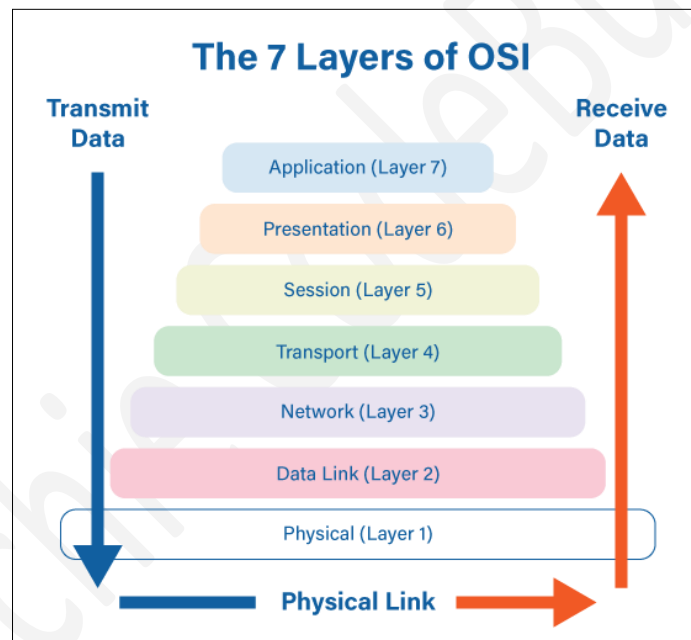
- Define responsibilities of different networking components.
- Help different technologies and protocols work together.
- Provide a common framework for understanding how data moves from one device to another.

Networking Model = Layered framework for network communication

##OSI Model

OSI (Open Systems Interconnection) is a **7-layer reference model** developed by the **International Organization for Standardization (ISO)**.

It provides a standardized way to understand how network communication functions from the physical transmission of data up to applications used by end users.

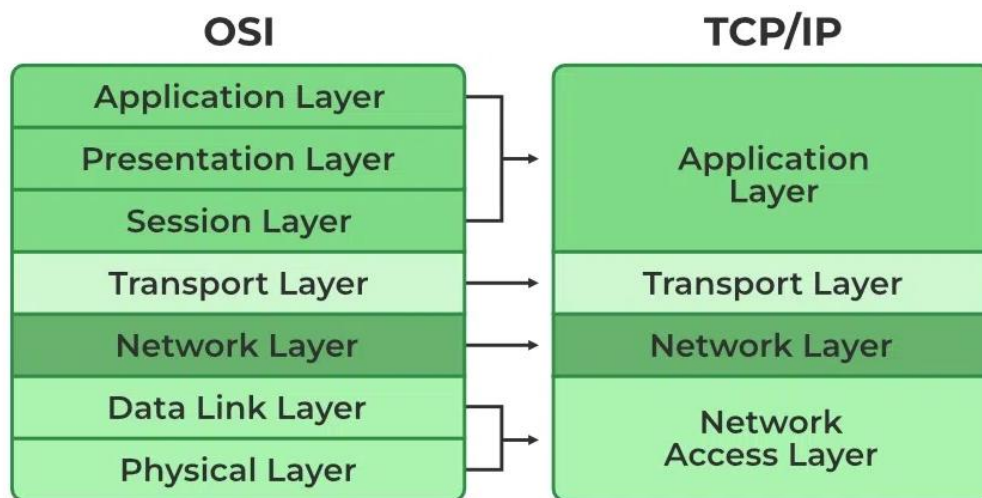


TCP/IP Model

The **TCP/IP model** is the practical networking model underlying the Internet and the TCP/IP protocol suite.

Layers

1. Network Access Layer
2. Internet Layer
3. Transport Layer
4. Application Layer



✂ OSI vs TCP/IP

Feature	OSI Model	TCP/IP Model
Full Form	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Developed by	ISO	DARPA/DoD research community
Number of Layers	7	4 commonly
Nature	Reference/conceptual model	Practical protocol suite/model
Session & Presentation	Separate layers	Included within Application
Network Access	Physical + Data Link are separate	Combined into Network Access
Transport Layer	Present	Present
Network/Internet Layer	Network	Internet
Common Protocols	Model itself is protocol-independent	TCP/IP protocol suite
Use	Understanding, design, standardization and troubleshooting	Internet and real-world networking

4.2 OSI Model — 7 Layers

The **OSI (Open Systems Interconnection) Model** divides network communication into **7 layers**. Each layer performs a specific set of functions and communicates with the layer directly above and below it.

OSI Layers — Top to Bottom

7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Data Link
1. Physical

Memory Trick

All People Seem To Need Data Processing

Layer 1 — Physical Layer

The **Physical Layer** is responsible for the **actual transmission of raw bits (0s and 1s) over a physical medium**.

What does it do?

- Converts bits into physical signals such as **electrical, optical or radio signals**.
- Deals with **bit transmission**, data rate and physical network characteristics.
- It does **not understand IP addresses, MAC addresses or frames**.
- Concerned with how bits are physically represented and transmitted.

Data Considered as: Bits

Devices

- Hub
- Repeater
- Cables/connectors

- Physical transceivers

A modem's signal-conversion functions can involve the physical layer, although a modem is not strictly limited to a single OSI layer.

Examples / Technologies

- Ethernet physical standards
- Fiber-optic transmission
- Copper cables
- Radio transmission
- Physical connectors

Example

When a computer sends:

10110101

the Physical Layer is concerned with **turning those bits into signals and transmitting them through the medium.**

Common Trap

Physical Layer does NOT deal with MAC addresses.

MAC addressing belongs primarily to **Layer 2**.

Remember

L1 → Bits → Signals → Physical Medium

Layer 2 — Data Link Layer

The **Data Link Layer** provides **node-to-node/link-level communication** and organizes network-layer packets into frames.

What does it do?

- Encapsulates Network Layer packets into **frames**.
- Uses **MAC addresses** for link-layer addressing in technologies such as Ethernet.

- Provides **frame delivery over a local/link network**.
- Performs error detection using mechanisms such as frame checks.
- Controls access to the shared communication medium in technologies where this is required.
- Helps switches make forwarding decisions in Ethernet networks.

Data Considered as Frames.

Address

MAC Address

Devices

- **Switch**
- **Bridge**

Important Technologies / Protocols

- Ethernet
- Wi-Fi (IEEE 802.11)
- PPP

Example

Suppose a packet needs to travel from your laptop to your **local router**.

The Data Link Layer encapsulates the packet into a frame containing link-layer addressing information.

Common Trap

MAC Address → Layer 2

IP Address → Layer 3

Remember

L2 → Frame → MAC → Local/Link Delivery

Layer 3 — Network Layer

The **Network Layer** is responsible for **logical addressing and routing packets between different networks**.

This is one of the most important layers for MCQs.

What does it do?

- Provides **logical addressing** using IP addresses.
- Determines how packets should travel between networks.
- Routers use Layer 3 information to make forwarding decisions.
- Handles packet forwarding based on the destination address.
- Supports routing and internetwork communication.
- The Network Layer is concerned with **where the packet needs to go**, not the physical link it is currently using.

Data Considered as Packet

Address

IP Address

Device

Router

Important Protocols

- **IPv4**
- **IPv6**
- **ICMP**
- IP-related routing mechanisms

Example

Your laptop wants to communicate with a server on another network:

Laptop → Router → ISP → Destination Network

The Network Layer uses the **destination IP address** to help determine where the packet should be forwarded.

Common Trap

Router → Layer 3

But remember that modern routers can perform functions at multiple layers. For basic OSI questions, **Router = Network Layer** is the standard answer.

Remember

L3 → Packet → IP → Routing

Layer 4 — Transport Layer

The **Transport Layer** provides **end-to-end communication between applications/processes** running on different hosts.

This is another **very high-value layer**.

What does it do?

- Provides communication between applications on different devices.
- Uses **port numbers** to identify the destination application/process.
- Can provide reliable, ordered delivery depending on the protocol.
- Handles mechanisms such as **segmentation, flow control and error recovery** in protocols such as TCP.
- TCP and UDP are the major transport protocols.
- Allows multiple applications to communicate simultaneously using different port numbers.

Data Considered as Segment/Datagram

TCP → Segment

UDP → Datagram

Address / Identifier

Port Number

Important Protocols

- TCP
- UDP

Example

Your computer may simultaneously use:

- Browser → Port 443
- DNS application → Port 53
- Other applications → Other ports

The Transport Layer helps deliver data to the **correct application/process**.

TCP vs UDP

TCP

- Connection-oriented
- Reliable
- Ordered
- Retransmission
- Flow/congestion control

UDP

- Connectionless
- No delivery guarantee
- Lower overhead
- No TCP-style retransmission

Common Trap

Port Number → Transport Layer

Don't confuse:

IP → Layer 3

Port → Layer 4

Remember

L4 → Segment/Datagram → Port → End-to-End

Layer 5 — Session Layer

The **Session Layer** is responsible for **establishing, managing and terminating communication sessions** between applications.

What does it do?

- Establishes a communication session.
- Maintains the session while communication is taking place.
- Manages dialog between communicating applications.
- Can provide synchronization/checkpoint mechanisms in appropriate systems.
- Helps coordinate when communication sessions start, continue and end.

The data passed between layers is generally referred to as **data** at these upper layers.

Address

No specific addressing scheme.

Device

No dedicated common networking device is associated with Layer 5.

Simple Example

Imagine a long file transfer or communication session that supports **checkpoints**.

If communication is interrupted, synchronization mechanisms can help resume from an appropriate point rather than treating the entire interaction as a completely new session.

Important Reality

In the practical TCP/IP model, **Session Layer functions are generally incorporated into the Application Layer** rather than existing as an independent protocol layer.

Remember

L5 → Session → Establish + Manage + Terminate

Layer 6 — Presentation Layer

The **Presentation Layer** deals with the **representation and transformation of data** so that communicating systems can interpret it correctly.

Think:

“How should the data be represented?”

What does it do?

Commonly associated with:

- **Data translation / format conversion**
- **Encryption and decryption**
- **Compression and decompression**
- Character encoding and data representation

Examples

Different systems may represent data differently. The Presentation Layer conceptually ensures that data can be represented in a form the receiving system can understand.

Address

No specific addressing scheme.

Device

No dedicated common networking device.

Example

Conceptually:

Original Data → Encryption/Encoding/Compression → Transmission

and at the receiving side:

Decompression/Decryption/Translation → Usable Data

Layer 7 — Application Layer

The **Application Layer** is the **top layer of the OSI model** and provides network services that applications use to communicate.

What does it do?

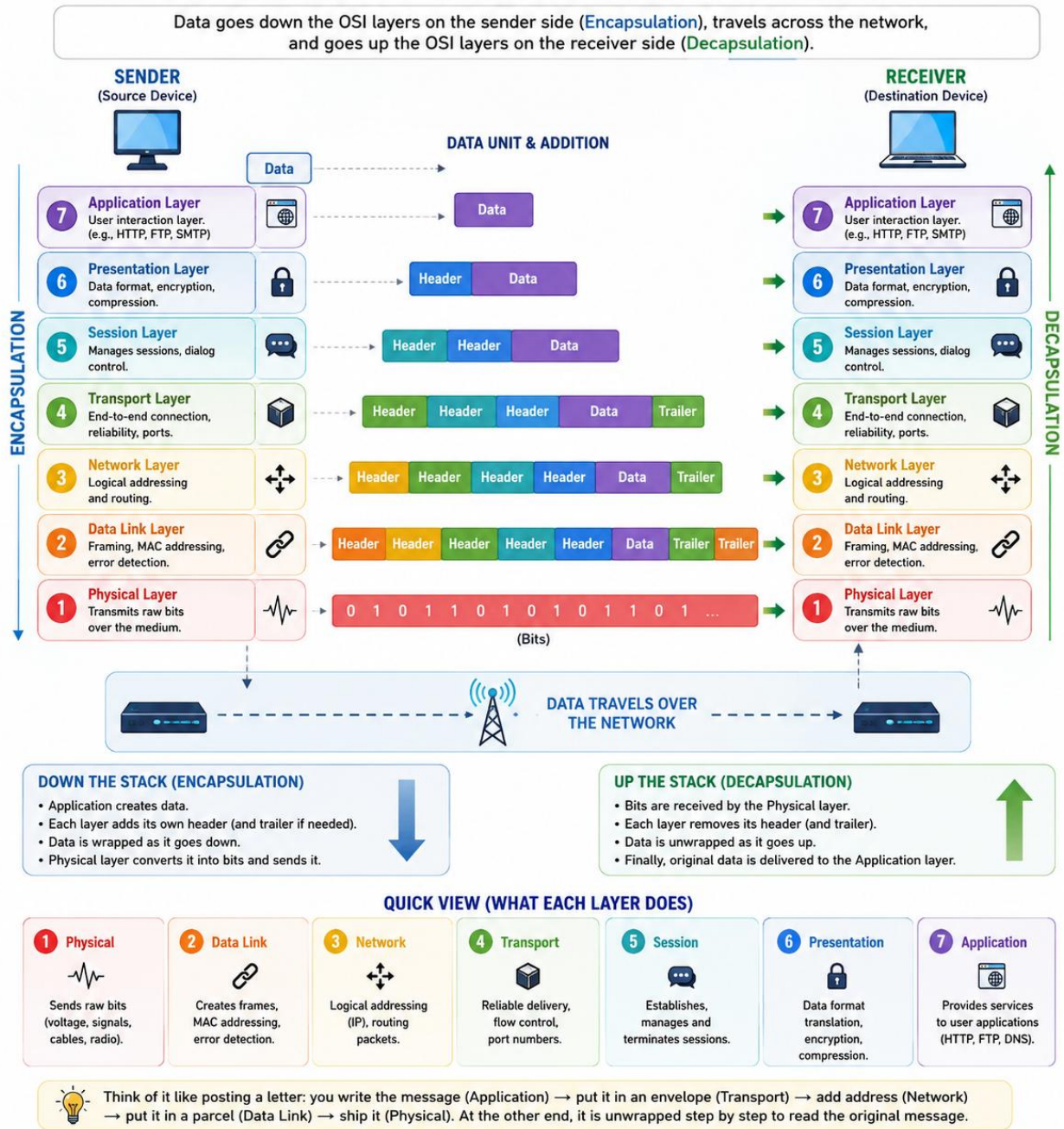
- Provides network-related services to application software.
- Enables applications to access network resources and services.
- Defines application-level communication protocols.
- Handles services such as web access, email, file transfer and name resolution.
- It is the layer **closest to the end user/application**.

Network Models



TCP/IP MODEL	OSI MODEL	PROTOCOLS
Application Layer	Application Layer	FTP,HTTP,Telnet
	Presentation Layer	JPEG,MPEG
	Session Layer	NFS,SQL,PAP
Transport Layer	Transport Layer	TCP,UDP
Network Layer	Network Layer	IPv4,IPv6
Network Access Layer	Data Link Layer	ARP,CDP,STP
	Physical Layer	Ethernet,Wi-Fi

HOW DATA MOVES THROUGH THE OSI LAYERS



4.4 TCP/IP Model

The **TCP/IP Model** is a practical networking model used to describe how data is communicated across the Internet and other IP-based networks.

It is commonly represented using **4 layers**:

4. Application
3. Transport
2. Internet

1. Network Access Layer

The **Network Access Layer** is responsible for transferring data over the **local network/link and physical medium**.

Main Functions

- Handles communication between devices on the **same local network/link**.
- Deals with **framing and link-level addressing**.
- Uses technologies such as **Ethernet and Wi-Fi**.
- Handles access to the physical transmission medium.
- Covers functions corresponding broadly to the **OSI Data Link + Physical layers**.
- Data is transmitted as **frames and ultimately bits/signals**.

Examples

- Ethernet
- Wi-Fi
- PPP

Devices commonly associated

- Switch
- Bridge
- Hub
- Repeater

Network Access = Local Link + Frames + Physical Transmission

2. Internet Layer

The **Internet Layer** is responsible for **logical addressing and routing packets between networks**.

Main Functions

- Provides **IP addressing**.
- Determines how packets are forwarded between different networks.
- Handles packet delivery across interconnected networks.
- Uses routers to forward packets toward their destinations.
- Provides the foundation for **internetwork communication**.
- Corresponds broadly to the **OSI Network Layer**.

Main Protocols

- **IPv4**
- **IPv6**
- **ICMP**

PDU: Packet

Address: IP Address

Device: Router

Internet Layer = IP + Routing + Packets

3. Transport Layer

The **Transport Layer** provides **end-to-end communication between applications/processes** running on different hosts.

Main Functions

- Provides communication between applications on different devices.
- Uses **port numbers** to identify application/process endpoints.
- Can provide reliable, ordered delivery through **TCP**.
- **UDP** provides lightweight, connectionless communication without TCP-style delivery guarantees.
- TCP also provides mechanisms such as **flow control and congestion control**.
- Corresponds broadly to the **OSI Transport Layer**.

Main Protocols

TCP — Transmission Control Protocol

UDP — User Datagram Protocol

PDU

- TCP → **Segment**
- UDP → **Datagram**

Identifier: Port Number

Transport = End-to-End + Ports + TCP/UDP

4. Application Layer

The **Application Layer** provides network services and protocols used directly by applications.

It combines the functions of the **OSI Application, Presentation and Session layers**.

Main Functions

- Provides network services to application software.
- Defines application-level communication protocols.
- Handles services such as web communication, email, file transfer and name resolution.
- Includes protocols required for applications to communicate over TCP/IP networks.
- Covers functions such as data representation and session management that are separate layers in OSI.
- It is the **top layer** of the TCP/IP model.

5. Cloud Computing – Introduction

5.1 Introduction to Cloud Computing

Cloud Computing is the delivery of computing resources and services—such as servers, storage, databases, networking and software—over a network (typically the Internet), on demand.

Instead of owning and maintaining all computing infrastructure locally, users can access resources provided by a cloud provider.

Key Points

- Resources are available **on demand**.
- Services can be accessed remotely through a network.
- Users can scale resources according to their requirements.
- Cloud providers manage much of the underlying infrastructure.
- Resources can be shared among multiple customers through **resource pooling**.
- Typically follows a **pay-as-you-use / usage-based** model, depending on the service.

Example

Instead of purchasing and maintaining your own physical server:

Traditional → Buy Server → Install → Maintain

With cloud:

User → Internet → Cloud Provider → Required Resources

 **Remember**

Cloud Computing = On-demand access to shared computing resources over a network

Why Cloud Computing?

Cloud computing is used because organizations often need computing resources without having to purchase and maintain all the underlying infrastructure themselves.

Major Reasons

1. Cost Efficiency

Avoids or reduces large upfront investment in physical infrastructure.

2. Scalability

Resources can be increased or decreased according to demand.

3. Flexibility

Users can access computing resources from different locations and devices.

4. Faster Deployment

Servers, storage and other resources can often be provisioned much faster than purchasing and setting up physical infrastructure.

5. Resource Utilization

Cloud providers can pool infrastructure and allocate resources dynamically among customers.

6. Reliability & Availability

Cloud platforms can use redundant infrastructure and multiple locations to improve availability.

7. Maintenance Reduction

The provider handles much of the underlying hardware and infrastructure maintenance, depending on the service model.

8. Disaster Recovery & Backup

Cloud infrastructure can support geographically distributed backup and recovery strategies.



Traditional Computing vs Cloud Computing

Feature	Traditional Computing	Cloud Computing
Infrastructure	Organization owns/controls local infrastructure	Resources provided by cloud provider
Initial Cost	Usually high	Lower upfront infrastructure cost
Scaling	Requires purchasing/configuring additional hardware	Resources can generally be scaled on demand
Provisioning	Often slower	Usually faster

Maintenance	Organization manages physical infrastructure	Provider manages much of the underlying infrastructure
Accessibility	Often dependent on local infrastructure/network	Resources accessible over a network
Resource Usage	Fixed/local capacity	Shared/pool-based resources
Payment	Hardware purchase + maintenance	Often usage/subscription based
Disaster Recovery	Organization must build/manage it	Cloud services can provide distributed backup/recovery options

Cloud Infrastructure —

Cloud Infrastructure refers to the underlying **physical and virtual resources** that enable cloud services.

Think of it as:

Physical Hardware + Virtualization + Networking + Storage + Management

1. Data Centers

Physical facilities where cloud providers host their computing infrastructure.

They contain:

- Servers
- Storage systems
- Networking equipment
- Power and cooling systems
- Security and monitoring systems

2. Servers / Compute

Physical servers provide the **processing power** needed to run applications and services.

Cloud providers can divide physical computing resources into virtual resources that customers can use.

3. Storage

Cloud infrastructure provides different forms of data storage for:

- Files
- Databases
- Backups
- Application data

Cloud storage allows data to be accessed over a network rather than being limited to a local physical disk.

4. Networking

Networking connects:

Users ↔ Cloud Resources ↔ Internet ↔ Data Centers

It includes components and technologies such as:

- Routers
- Switches
- Load balancers
- Virtual networks
- Firewalls

5. Virtualization

Virtualization allows physical computing resources to be divided into multiple **virtual resources**.

For example:

Physical Server → Virtual Machine 1 + Virtual Machine 2 + Virtual Machine 3

This helps cloud providers utilize physical infrastructure efficiently and provide isolated computing environments to different users.

6. Management & Orchestration

Cloud platforms use software to **provision, monitor, configure and manage resources**.

For example, a user can request a virtual machine through a cloud console/API without physically accessing the server.

5.2 Characteristics of Cloud Computing

Cloud computing has several characteristics that distinguish it from traditional computing. These characteristics explain **how cloud resources are provided, accessed, shared and scaled**.

1. ⚡ On-Demand Self-Service

On-Demand Self-Service means users can **provision computing resources automatically whenever they need them**, without requiring direct human interaction with the service provider.

Key Points

- Users can request resources such as **VMs, storage or databases** on demand.
- Provisioning can be done through a **web console, API or other automated interface**.
- Users do not need to physically contact the provider to allocate resources.
- Resources can usually be released when they are no longer required.

Example

A developer creates a virtual machine through a cloud console within minutes instead of asking an administrator to install a physical server.

Remember → User needs resource → User provisions it → No manual provider intervention

2. Broad Network Access

Broad Network Access means cloud capabilities are available over a network and can be accessed through **standard mechanisms from a variety of devices**.

Key Points

- Cloud services are accessed through a network, commonly the Internet.
- Can be accessed from devices such as **laptops, smartphones, tablets and workstations**.
- Uses standard network protocols and interfaces.
- Enables users to access resources from different locations.

Example

A user accesses **Google Drive** from a laptop, smartphone or tablet through an Internet connection.

Remember → Cloud services + Network + Multiple device types

3. Resource Pooling

Resource Pooling means the cloud provider's physical and virtual computing resources are **pooled together to serve multiple customers**.

Key Points

- Provider maintains a shared pool of **compute, storage and networking resources**.
- Resources are dynamically allocated according to customer requirements.
- Physical resources are not necessarily dedicated to one customer.
- Users generally have limited knowledge/control over the exact physical location of resources.
- Supports efficient utilization of provider infrastructure.

Example

A cloud provider has a large pool of servers and dynamically allocates computing capacity to different customers according to demand.

Remember → Shared resource pool → Dynamic allocation

4. 📐 Scalability

Scalability is the ability of a system to **increase or decrease its resource capacity to handle changes in workload**.

Key Points

- Helps systems handle increasing demand.
- Resources can be added or removed as requirements change.
- Can involve **vertical scaling** or **horizontal scaling**.

Vertical Scaling

Increase the resources of an existing machine.

Example:

4 GB RAM → 16 GB RAM

Horizontal Scaling

Add more machines/instances.

Example:

2 Servers → 5 Servers

Remember → Scalability = Ability to handle changing workload by adjusting capacity

5. ● Availability

Availability refers to the ability of a cloud service or system to **remain accessible and operational when users need it**.

Key Points

- High availability aims to minimize service downtime.

- Cloud providers can use **redundancy and fault-tolerant infrastructure** to improve availability.
- Multiple servers, network paths or data-center locations may be used depending on the architecture.
- Availability is often expressed as a percentage, such as **99.9% uptime**.
- High availability does **not mean zero downtime**.

Example

If one server fails, traffic can be redirected to another healthy server so that users can continue accessing the application.

Remember → Availability = Service remains accessible

5.3 Benefits of Cloud Computing

Benefit	Remember
Cost Reduction	Lower infrastructure investment
Scalability	Adjust capacity
Flexibility	Adapt resources quickly
Accessibility	Access remotely
Reliability	Redundancy & fault tolerance
Collaboration	Shared resources
Disaster Recovery	Recover after disruption
Business Continuity	Continue operations

5.4 Cloud Deployment Models

A cloud deployment model describes how cloud infrastructure is organized, owned, managed, and made available to users.

The four commonly discussed deployment models are:

Public → Private → Hybrid → Community

1. Public Cloud

A **Public Cloud** is a cloud environment where computing resources are provided by a **third-party cloud provider** and made available to multiple customers over a network.

Key Points

- Infrastructure is owned/operated by a **cloud service provider**.
- Resources are shared among multiple customers, with **logical isolation** between tenants.
- Customers generally pay according to their selected service/usage.
- Offers high scalability and easy provisioning.
- Suitable for organizations that don't want to maintain their own physical cloud infrastructure.

Examples

AWS, Microsoft Azure, Google Cloud

Remember

Public Cloud = Provider-owned + Shared infrastructure + Multiple customers

2. Private Cloud

A **Private Cloud** is a cloud environment dedicated to a **single organization**.

Key Points

- Infrastructure is dedicated to one organization.
- Can be **owned, managed and operated by the organization or a third party**.
- Provides greater control over infrastructure, configuration and security policies.
- Can be deployed **on-premises or hosted externally**.

- Generally involves more responsibility and cost than using a public cloud.

Example

A bank operates a cloud environment dedicated exclusively to its own applications and data.

Remember

Private Cloud = One organization + Dedicated environment

3. Hybrid Cloud

A **Hybrid Cloud** combines **two or more distinct cloud/infrastructure environments**—typically private and public cloud—with mechanisms that allow them to work together or exchange data/applications.

Key Points

- Combines **private + public cloud** environments.
- Allows organizations to keep certain workloads/data in a private environment while using public cloud resources for others.
- Can provide greater flexibility and scalability.
- Useful for **workload portability, cloud bursting and gradual cloud migration**.
- Requires appropriate integration, networking and security mechanisms.

Example

A company keeps sensitive systems in its **private cloud** but uses **public cloud** resources during periods of high demand.

Remember

Hybrid Cloud = Combination + Integration

4. Community Cloud

A **Community Cloud** is a cloud infrastructure designed for use by a **specific community of organizations** that share common requirements, such as security, compliance, mission or policy needs.

Key Points

- Used by a **specific group of organizations**.
- Organizations share common concerns or requirements.
- Infrastructure can be **owned, managed and operated by one or more organizations in the community, a third party, or a combination**.
- Can be hosted on-premises or externally.
- Useful when organizations have common regulatory, security or operational requirements.

Example

Several government organizations with similar security and compliance requirements share a dedicated cloud environment.

Remember

Community Cloud = Shared by organizations with common requirement

 Deployment Model Comparison 				
Feature	Public	Private	Hybrid	Community
Users	Multiple customers	Single organization	Combination of environments	Specific community
Infrastructure	Provider-owned/operated	Dedicated to one organization	Multiple environments	Shared by community
Control	Lower	Higher	High for private part	Shared/community governance
Scalability	★ High	Depends on infrastructure	★ High	Depends on design
Cost	Generally lower upfront	Generally higher	Variable	Shared among participants
Main Advantage	Easy access & scalability	Control & customization	Flexibility	Common requirements
Typical Use	General-purpose cloud services	Sensitive/controlled workloads	Mixed workloads	Organizations with common needs

5.6 Major Cloud Providers

The three major cloud providers commonly encountered in cloud-computing questions are **Amazon Web Services (AWS)**, **Microsoft Azure**, and **Google Cloud**.

The important thing for a revision cheatsheet is **not memorizing hundreds of services**, but knowing the major services and their equivalents.

1. AWS — Amazon Web Services

AWS is the cloud platform provided by **Amazon**.

It offers services for computing, storage, databases, networking, security, analytics and more.

Important AWS Services

Service	Purpose
EC2	Virtual servers / compute
S3	Object storage
RDS	Managed relational databases
VPC	Isolated virtual network
Lambda	Serverless compute
CloudFront	Content delivery network (CDN)
IAM	Identity and access management

2. Microsoft Azure

Microsoft Azure is Microsoft's cloud computing platform.

It provides services for computing, storage, databases, networking, AI, security and enterprise applications.

Service	Purpose
Azure Virtual Machines	Virtual servers / compute
Azure Blob Storage	Object storage
Azure SQL Database	Managed relational database
Azure Virtual Network (VNet)	Virtual networking
Azure Functions	Serverless compute
Azure CDN	Content delivery
Microsoft Entra ID	Identity and access management

3. Google Cloud

Google Cloud is Google's cloud computing platform.

It provides cloud services for computing, storage, databases, networking, analytics, AI and more.

Important Google Cloud Services

Service	Purpose
Compute Engine	Virtual machines / compute
Cloud Storage	Object storage
Cloud SQL	Managed relational database
VPC	Virtual networking
Cloud Functions	Serverless compute
Cloud CDN	Content delivery
Cloud IAM	Identity and access management

6. Cloud Service Models

6.1 IaaS —

IaaS (Infrastructure as a Service) is a cloud service model in which the cloud provider offers **virtualized computing infrastructure**—such as virtual machines, storage and networking—over a network.

The customer gets control over the operating system and applications without having to purchase and maintain the underlying physical hardware.

Definition & Key Features

What does IaaS provide?

- **Compute** → Virtual machines / processing resources
- **Storage** → Cloud-based storage
- **Networking** → Virtual networks, IPs, firewalls, etc.
- **Virtualization** → Resources are provided as virtual infrastructure.
- **On-demand provisioning** → Resources can be created/released as needed.
- **Scalability** → Resources can generally be increased or decreased according to workload.

Simple Example

Instead of buying a physical server:

Traditional:

Buy Server → Install OS → Configure → Maintain Hardware

IaaS:

Choose VM → Select Resources → Deploy → Install/Manage OS → Run Applications

IaaS = Rent/Provision Infrastructure instead of owning the physical hardware

IaaS Examples

AWS EC2

Amazon EC2 (Elastic Compute Cloud) provides **resizable virtual computing capacity** in AWS.

EC2 → Virtual Machines / Compute

Microsoft Azure Virtual Machines

Azure Virtual Machines provides scalable virtual machines that allow users to run operating systems and applications in Azure.

Azure VM → Virtual Machines / Compute

Google Compute Engine

Google Compute Engine provides configurable virtual machines running on Google's infrastructure.

Compute Engine → Virtual Machines / Compute

6.2 PaaS — Platform as a Service

PaaS (Platform as a Service) is a cloud service model that provides a **ready-to-use application development and deployment environment**. The cloud provider manages the underlying infrastructure and platform components, allowing developers to focus mainly on their **application code and data**.

PaaS = Build & Deploy Applications without managing the underlying infrastructure

Definition & Key Features

What does PaaS provide?

- Provides a **development and deployment platform** over the cloud.
- Provider manages servers, storage, networking and much of the underlying infrastructure.
- Provides platform components such as **operating system, runtime and middleware**.
- Developers can focus on **writing, testing and deploying applications**.
- Reduces the need for infrastructure setup and maintenance.
- Supports faster application development and deployment.

Simple Example

Instead of:

Buy Server → Install OS → Configure Runtime → Install Middleware → Deploy App

With PaaS:

Choose Platform → Write Code → Deploy Application

PaaS = Developer focuses on application; provider manages the platform

6.3 SaaS — Software as a Service

SaaS (Software as a Service) is a cloud service model in which **complete software applications are provided to users over a network**, usually through a web browser or app.

The user consumes the software as a service instead of installing, maintaining, and managing the underlying infrastructure and application platform.

SaaS = Ready-to-use software

Definition & Key Features

What does SaaS provide?

- Provides a **complete, ready-to-use application**.
- Usually accessed through a **web browser or dedicated application**.
- The provider manages the underlying **servers, storage, networking, OS, platform and application**.
- Users generally don't need to install or maintain the underlying software infrastructure.
- Updates, patches and maintenance are largely handled by the provider.
- Usually offered through a **subscription or usage-based model**, depending on the service.

Simple Example

Instead of:

Install Software → Configure → Update → Maintain

With SaaS:

Open Application → Sign In → Use

SaaS = Use the software; provider manages the rest

SaaS Examples

1. Gmail

Gmail is a cloud-based email service that allows users to **send, receive and manage email through the Internet**.

The user does not manage the underlying mail servers or software infrastructure.

Gmail → SaaS

2. Google Workspace

Google Workspace provides cloud-based productivity and collaboration applications such as:

- Gmail
- Google Docs
- Google Sheets
- Google Drive
- Google Meet

Users access these applications as services rather than managing the underlying infrastructure.

Google Workspace → SaaS

3. Microsoft 365

Microsoft 365 is a subscription-based suite of cloud productivity applications and services.

Examples include:

- Word
- Excel
- PowerPoint
- Outlook
- Teams
- OneDrive

The underlying infrastructure and service platform are managed by Microsoft.

Microsoft 365 → SaaS

4. Dropbox

Dropbox provides cloud-based file storage, synchronization and sharing.

Users can store and access their files through the Internet without managing the underlying storage infrastructure.

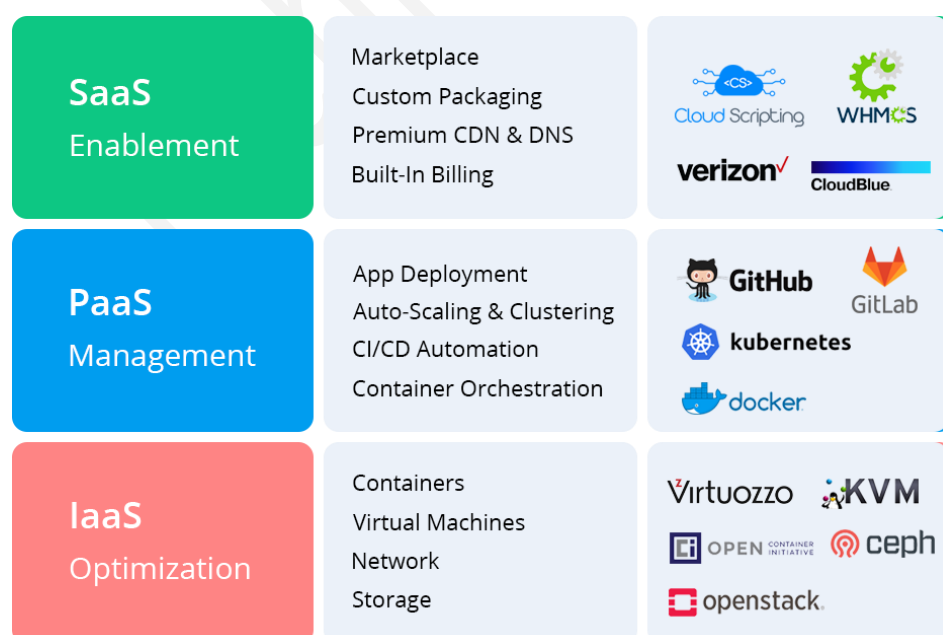
Dropbox → SaaS

5. Canva

Canva is a cloud-based design and visual-content creation platform.

Users can create and edit designs through the service without managing the underlying servers, operating systems or application infrastructure.

Service	What the User Gets	Model
Gmail	Email service	SaaS
Google Workspace	Productivity & collaboration apps	SaaS
Microsoft 365	Productivity applications/services	SaaS
Dropbox	Cloud file storage & synchronization	SaaS
Canva	Online design application	SaaS



✂ IaaS vs PaaS vs SaaS

This is the **most important revision concept** of the entire Cloud Service Models section.

	IaaS	PaaS	SaaS
What you get	Infrastructure	Development platform	Complete software
User manages	OS + Apps + Data	Application + Data	Mainly usage/configuration + data
Provider manages	Physical infrastructure	Infrastructure + OS + Runtime	Almost entire stack
Control	★ Highest	Medium	Lowest
User effort	Highest	Medium	Lowest
Example	AWS EC2	Google App Engine	Gmail
Think	Build infrastructure	Build application	Use application

🧠 IaaS vs PaaS vs SaaS

Feature	IaaS	PaaS	SaaS
Full Form	Infrastructure as a Service	Platform as a Service	Software as a Service
What is provided?	Virtual infrastructure	Application development/deployment platform	Complete software application
User Control	Highest	Medium	Lowest
User mainly manages	OS, applications, data	Application/code, data & configuration	Application usage/configuration & data
Provider manages	Physical infrastructure + virtualization	Infrastructure + OS + runtime/platform	Almost entire technology stack
Infrastructure knowledge required	High	Medium	Low
Best suited for	IT admins / system engineers	Developers	End users/business users
Example	AWS EC2	Google App Engine	Gmail
Easy thought	"Give me a machine."	"Give me a platform."	"Give me software."

7. Cryptography & Encryption Algorithms

7.1 Cryptography Basics

Cryptography is the foundation of many security mechanisms used to **protect information from unauthorized access or modification**.

The basic flow is:

Plaintext → Encryption + Key → Ciphertext → Decryption + Key → Plaintext

What is Cryptography?

Cryptography is the science and practice of using **mathematical techniques, algorithms and keys to protect information** and secure communication.

It can help provide security properties such as:

- **Confidentiality** → prevents unauthorized disclosure
- **Integrity** → helps detect unauthorized modification
- **Authentication** → helps verify identity/source
- **Non-repudiation** → supports proof of origin in appropriate systems

Example

Suppose you want to send:

MEET AT 5 PM

Cryptography can transform this understandable information into an unintelligible form before transmission, and the authorized recipient can recover or verify the information using the appropriate cryptographic mechanism.

Cryptography = Protect information using algorithms + keys

1. Plaintext

Plaintext is the **original, readable or understandable information** before encryption.

It can be:

- Text, Numbers, Images, Files, Messages or any other Digital Data

Example

Plaintext:

MEET AT 5 PM

Plaintext = Original/understandable data

 **Remember**

Plain = Readable

2. Ciphertext

Ciphertext is the **transformed, unintelligible form of data produced by encryption**.

Without the appropriate cryptographic key and algorithm, an unauthorized party should not be able to recover the protected plaintext from ciphertext when a secure encryption scheme is properly used.

Example

Plaintext

↓ Encryption

Ciphertext

The ciphertext may look like random characters or binary data.

Ciphertext = Encrypted/protected data

 **Remember**

Cipher = Scrambled/encoded-looking result

3. Encryption

Encryption is the process of transforming **plaintext into ciphertext** using a cryptographic algorithm and a key.

Basic Flow

Plaintext

+

Key

↓

Encryption Algorithm

↓

Ciphertext

Purpose

Primarily provides:

Confidentiality

It makes the information unreadable to parties who do not have the required means to decrypt it.

Example

Sender:

"HELLO"

↓ Encryption

Ciphertext

The ciphertext is transmitted to the receiver.

Encryption = Plaintext → Ciphertext

4. Decryption

Decryption is the process of recovering the **plaintext** from **ciphertext** using the appropriate key and cryptographic algorithm.

Basic Flow

Ciphertext

+

Key

↓

Decryption Algorithm



Plaintext

Example

Receiver:

Ciphertext

↓ Decryption

"HELLO"

Decryption = Ciphertext → Plaintext

5. Key

A **cryptographic key** is a value used by a cryptographic algorithm to **control the transformation of data** during encryption, decryption, signing or verification, depending on the cryptographic operation.

Think of the key as an important **secret or controlled parameter** that works with the algorithm.

Important Points

- Different cryptographic systems use different types and sizes of keys.
- In **symmetric cryptography**, the same shared secret key is generally used for encryption and decryption.
- In **asymmetric cryptography**, a related **public/private key pair** is used.
- Security depends not only on the algorithm but also on **proper key management and protection**.

Simple Analogy

Algorithm = Lock mechanism

Key = Key used with that mechanism

 **Important**

A cryptographic key is **not the same thing as a password**, although passwords can sometimes be used to derive cryptographic keys through specialized key-derivation functions.

⚙️ 6. Cryptographic Algorithm

A **cryptographic algorithm** is a **mathematical procedure** used to perform a cryptographic operation such as encryption, decryption, hashing or digital signatures.

Examples include:

- **AES** → Symmetric encryption
- **RSA** → Asymmetric cryptography
- **ECC** → Public-key cryptography

Important

The **algorithm and key work together**.

Algorithm = Method

Key = Value that controls the operation

A strong cryptographic system should not depend on keeping the algorithm secret; security should primarily rely on **properly protecting the key**.

🧠 MASTER REVISION TABLE

Term	Meaning	Remember
Cryptography	Techniques used to protect information using cryptographic methods	Information Security
Plaintext	Original understandable data	Readable
Ciphertext	Transformed data produced by encryption	Protected/Unreadable
Encryption	Plaintext → Ciphertext	Lock/Protect
Decryption	Ciphertext → Plaintext	Recover/Unlock
Key	Value used with a cryptographic algorithm	Controls operation
Cryptographic Algorithm	Mathematical procedure used for a cryptographic operation	Method

7.2 Security Goals

Security goals describe **what a secure system should achieve**. The most fundamental goals are represented by the **CIA Triad**:

Confidentiality + Integrity + Availability

In addition, **Authentication** and **Non-Repudiation** are important security properties.

1. Confidentiality

Confidentiality ensures that information is **accessible only to authorized users or entities**.

Key Points

- Prevents unauthorized **disclosure or exposure** of information.
- Protects sensitive data from being read by unauthorized parties.
- Commonly supported using **encryption, access controls and authentication**.
- Particularly important for passwords, financial data and personal information.

Example

If an attacker intercepts an encrypted message but **cannot understand its contents**, confidentiality is being protected.

Confidentiality = Prevent unauthorized reading

2. Integrity

Integrity ensures that data remains **accurate, complete and unaltered by unauthorized parties**.

Key Points

- Protects data from unauthorized **modification, deletion or manipulation**.
- Helps ensure that received data is the same as the data that was originally sent/stored.

- Common mechanisms include **hashes, MACs and digital signatures**.
- Important for financial transactions, software, databases and messages.

Example

If a bank transaction says:

₹1,000 → ₹10,000

because an attacker modified the transaction, **integrity has been compromised**.

Integrity = Prevent unauthorized modification

3. Availability

Availability means authorized users should be able to **access systems, services and data when required**.

Key Points

- Focuses on keeping systems **operational and accessible**.
- Can be affected by hardware failures, network failures, outages or cyberattacks.
- Supported using **redundancy, backups, fault tolerance, load balancing and disaster recovery**.
- **DoS/DDoS attacks** specifically attempt to compromise availability.

Example

If an online banking website becomes unavailable because of a DDoS attack:

Availability is compromised.

Availability = Access when needed

4. Authentication

Authentication is the process of **verifying the identity of a user, device or entity**.

Authentication = “Who are you?”

Common Methods

- Password
- OTP
- Biometrics
- Security token
- Multi-factor authentication (MFA)

Example

You enter your username and password to log into your college portal.

The system verifies whether you are actually the claimed user.

Authentication → Identity Verification

5. Non-Repudiation

Non-repudiation provides evidence that helps establish that a particular entity **performed or authorized an action**, so that it cannot easily deny that action later.

Key Points

- Important for transactions and legally/significantly important communications.
- **Digital signatures** are a major mechanism used to provide non-repudiation in appropriate systems.
- Provides evidence of the **origin/authorship** and integrity of signed data.
- Strong non-repudiation depends on proper key management and the surrounding system/legal framework.

Example

A person digitally signs an electronic document.

The digital signature can provide evidence linking the signature to the corresponding private key and showing that the signed content has not been altered.

Non-Repudiation = Evidence against later denial

 Security Goals — Quick Comparison		
Goal	Main Question	Protects Against
Confidentiality	Who can read it?	Unauthorized disclosure
Integrity	Has it been changed ?	Unauthorized modification
Availability	Can I access it?	Downtime/disruption
Authentication	Who are you?	Impersonation
Non-Repudiation	Can you deny the action later?	Denial of origin/action

7.3 Symmetric Cryptography

Symmetric cryptography is a cryptographic approach in which the **same secret key** is used for both encryption and decryption.

Symmetric Cryptography = One shared secret key

7.4 Asymmetric Cryptography

Asymmetric cryptography (public-key cryptography) uses a **pair of mathematically related keys**: a **public key** and a **private key**.

Unlike symmetric cryptography, the same secret key is **not** used for both operations.

Asymmetric Cryptography = Public Key + Private Key

Symmetric vs Asymmetric

Feature	Symmetric Cryptography	Asymmetric Cryptography
Number of Keys	One shared secret key	Two related keys — Public + Private
Encryption / Decryption	Same secret key is used	Uses a related key pair
Speed	⚡ Fast	🐢 Generally slower
Computational Cost	Lower	Higher
Key Management	Key distribution can be difficult	Public key can be openly distributed; private key remains secret
Bulk Data Encryption	★ Highly suitable	Generally not suitable
Digital Signatures	✗ Not provided by encryption alone	★ Supported
Key Establishment	Requires secure key sharing/establishment	Can support secure key establishment
Typical Use	Encrypting files, storage, network traffic	Digital signatures, authentication, key establishment
Important Algorithms	AES, DES, 3DES, ChaCha20	RSA, ECC

Hashing

Hashing is a fundamental concept in cybersecurity, but it is important to understand one thing clearly:

Hashing ≠ Encryption

Hashing is primarily used to create a **fixed-length representation (hash value/digest) of data**, whereas encryption is designed to allow authorized parties to recover the original plaintext.

1. What is Hashing?

Hashing is the process of applying a **hash function** to input data to produce a fixed-size output called a **hash value** or **digest**.

✂️ 4. Hashing vs Encryption

This is a **very important Accenture distinction**.

Feature	Hashing	Encryption
Purpose	Integrity / verification	Confidentiality
Reversible?	Designed to be one-way	Yes, with appropriate key
Output	Fixed-length digest	Ciphertext generally related to input size
Key required?	Cryptographic hash functions don't require an encryption key	Requires key(s)
Original data recovery	❌ Not intended	✅ Possible with correct decryption key
Common use	Password verification, integrity	Secure data transmission/storage
Examples	SHA-256, SHA-3	AES, RSA

🧠 Easy Memory

Encryption → Lock the data → Unlock later

Hashing → Create a fingerprint → Verify later

7.7 Digital Signature

A **Digital Signature** is a cryptographic mechanism used to verify the **authenticity and integrity of digital data** and, in appropriate systems, provide evidence supporting **non-repudiation**.

It uses **asymmetric cryptography**:

Private Key → Sign

Public Key → Verify

⚠️ **Very Important**

A digital signature is **not the same as encryption**.

- **Encryption** → primarily provides **confidentiality**
- **Digital Signature** → provides **authentication + integrity + non-repudiation support**

1. What is a Digital Signature?

A digital signature is a cryptographic value generated by the **sender's private key** over the message/data (typically using a cryptographic hash of the data).

It allows the receiver to verify:

1. **Who signed it?** → Authentication
2. **Was the data modified?** → Integrity
3. **Can the signer reasonably deny signing it later?** → Non-repudiation support

Basic Concept

Message

↓

Hash Function

↓

Message Digest

↓

Sign using Sender's Private Key

↓

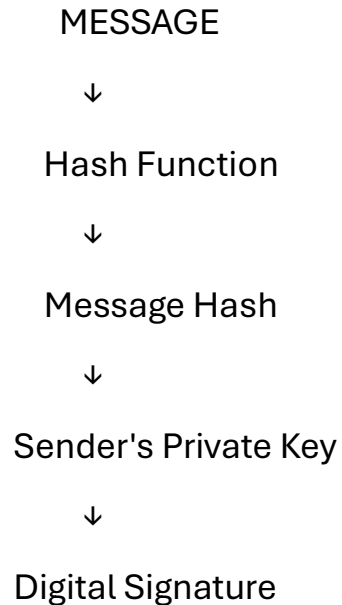
Digital Signature

The signature is normally sent along with the original message/data.

2. Signing

Signing is the process of generating a digital signature for data.

Sender Side



The sender sends:

Message + Digital Signature

Remember

Private Key → Signing

The private key must remain under the control of its owner.

3. Verification

Verification is the process of checking whether the digital signature is valid for the received data.

The receiver uses the **sender's public key** and the received data/signature.

Conceptually:

Received Message



Hash Function

↓

Hash A

Digital Signature

↓

Sender's Public Key

↓

Hash B / recovered signature value

Compare

↓

Match → Signature Valid

Mismatch → Invalid / Data or signature changed

If Verification Succeeds

It provides evidence that:

- The signature corresponds to the claimed public key.
- The signed data has not been altered since signing.
- The signer controlled the corresponding private key.

Public Key → Verification

⚙️ 4. How Digital Signature Works — Complete Flow

Let's say **Alice wants to digitally sign a document** and send it to Bob.

Step 1 — Alice has a key pair

Alice has:

- **Public Key** → Can be shared
- **Private Key** → Kept secret

Step 2 — Hash the document

Document



Hash Function



Hash Value

The hash produces a fixed-length digest of the document.

Step 3 — Create the signature

Alice uses her **private key** with the appropriate signature algorithm to create the digital signature over the document's hash.

Document Hash



Alice's Private Key



Digital Signature

Step 4 — Send

Alice sends:

Document + Digital Signature

to Bob.

Step 5 — Bob calculates the document hash

Bob independently hashes the received document:

Received Document



Hash Function

↓

Hash A

Step 6 — Bob verifies the signature

Bob uses **Alice's public key** with the appropriate verification algorithm.

Digital Signature

+

Alice's Public Key

↓

Verification

Bob then checks whether the signature is valid for the received document/hash.

Result

Valid → Signature corresponds to the data and public key.

Invalid → The signature does not correspond to the received data/key.

Authentication

Digital signatures can provide **authentication of the signer**, provided the public key has been reliably associated with that signer.

Why?

The signature was created using the signer's **private key**.

If the corresponding public key successfully verifies the signature, it provides evidence that the signature was created using the corresponding private key.

But there's an important condition:

How do we know the public key actually belongs to Alice?

This is where **Digital Certificates and Certificate Authorities (CAs)** become important.

Remember

Digital Signature → Authentication of signer

Certificate → Helps establish whose public key it is

Integrity

Digital signatures provide **integrity verification**.

If even a small part of the signed document changes:

Original Document

↓

Hash A

Modified Document

↓

Hash B

Hash A $\neq$ Hash B

The signature verification should fail because the signature no longer corresponds to the modified data.

Example

Alice signs:

Transfer ₹10,000

An attacker changes it to:

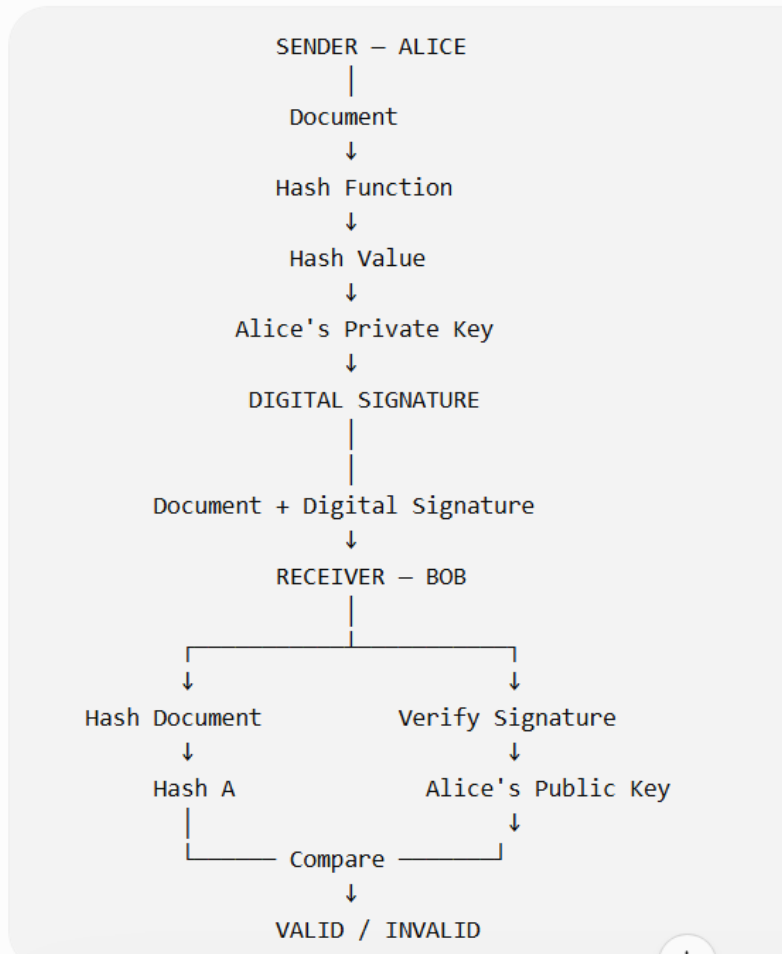
Transfer ₹90,000

The document's hash changes, causing signature verification to fail.

Digital Signature → Detects unauthorized modification



MASTER DIGITAL SIGNATURE FLOW



Digital Signature vs Encryption

This is a very common exam trap.

Feature	Digital Signature	Encryption
Main Purpose	Authentication + Integrity	Confidentiality
Sender's Private Key	Signing	Not normally used for ordinary confidentiality encryption
Recipient's Public Key	Not used for ordinary signature verification	Used to encrypt data for recipient
Verification	Sender's Public Key	Recipient decrypts using Private Key
Hides message?	✗ No	✓ Yes
Integrity	✓	Encryption alone does not necessarily provide integrity
Authentication	✓	Encryption alone does not necessarily authenticate sender
Non-Repudiation	Supports it	✗ Not by itself

7.8 Digital Certificate

A **Digital Certificate** is an electronic document used to **bind an identity to a public key**. It is issued and digitally signed by a trusted **Certificate Authority (CA)**.

Digital Certificate = Identity + Public Key + CA's Digital Signature

It helps users/devices verify that a particular public key actually belongs to the claimed entity.



1. Digital Certificate

A digital certificate contains information that associates a **public key with an identity**.

Typical Information

A certificate may contain:

- **Subject / Owner identity**
- **Subject's Public Key**
- **Certificate Issuer (CA)**
- **Validity period**
- **Certificate serial number**
- **Signature algorithm**
- **CA's Digital Signature**
- Other certificate extensions and constraints

Simple Example

Suppose you visit: <https://example.com>

Your browser receives a certificate claiming:

Identity: example.com

Public Key: ABC...

Issued by: Trusted CA

The browser can then verify the CA's signature and certificate details.

Certificate answers: “Whose public key is this?”

2. Certificate Authority (CA)

A **Certificate Authority (CA)** is a trusted entity that **issues and digitally signs certificates** after performing the required validation.

Main Responsibilities

- Verifies the identity/ownership information according to the certificate type and validation process.
- Issues digital certificates.
- Digitally signs certificates using the CA's private key.
- Helps establish trust between the certificate holder and relying parties.
- May revoke certificates when necessary.

Simple Flow

Entity

↓

Requests Certificate

↓

Certificate Authority (CA)

↓

Validates required information

↓

Issues + Signs Certificate

Important

The CA has its own **public/private key pair**.

CA Private Key → Signs Certificate

CA Public Key → Used to Verify CA Signature



Digital Certificate — Master Revision

Term	Meaning
Digital Certificate	Binds an identity to a public key
Public Key Certificate	Certificate containing identity/public-key information
CA	Trusted authority that issues/signs certificates
CA Private Key	Used to digitally sign certificates
CA Public Key	Used to verify CA signatures
Certificate Chain	Chain from server/leaf certificate through intermediates to trusted root
Validity Period	Defines when certificate is valid
CRL	List of revoked certificates
OCSP	Protocol for checking certificate status
TLS Certificate	Certificate used as part of TLS authentication

8. Cyber Attacks & Security Measures

8.1 Cybersecurity Fundamentals



1. Cybersecurity

Cybersecurity is the practice of protecting **systems, networks, applications and data** from unauthorized access, attacks, damage or disruption.

Cybersecurity = Protect digital systems & data



2. Information Security

Information Security (InfoSec) is the protection of **information and information systems** from unauthorized access, use, disclosure, modification, disruption or destruction.

Information Security = Protect information

3. Network Security

Network Security is the practice of protecting **network infrastructure, devices and network traffic** from unauthorized access, misuse and attacks.

Network Security = Protect networks & network traffic

▲ 4. CIA Triad

The **CIA Triad** is the three fundamental goals of information security:

- **Confidentiality** → Prevent unauthorized access/disclosure
- **Integrity** → Prevent unauthorized modification
- **Availability** → Keep systems/data accessible when required

CIA = Confidentiality + Integrity + Availability



5. Threat

A **threat** is a potential circumstance, event or actor that could **cause harm to a system, network or data**.

Threat = Potential source of harm

Example: Malware, attacker, natural disaster.

● 6. Vulnerability

A **vulnerability** is a **weakness or flaw** in a system, application, network or process that can potentially be exploited.

Vulnerability = Weakness

Example: Unpatched software vulnerability.

7. Risk

Risk is the potential for **loss or harm** resulting from a threat exploiting a vulnerability.

Risk = Likelihood × Potential Impact (*conceptually*)

Example: An unpatched server exposed to the Internet has a higher risk of being compromised.

8. Attack

An **attack** is an **intentional attempt to exploit a vulnerability** or compromise the confidentiality, integrity or availability of a system, network or data.

Attack = Actual attempt to cause harm/exploit weakness

Example: SQL Injection, DDoS, phishing.

8.2 Malware

Malware (Malicious Software) refers to software intentionally designed to **damage, disrupt, spy on, gain unauthorized access to, or otherwise compromise** systems, networks or data.

Malware = Malicious software designed to cause harm or unauthorized activity

1. Virus

A **virus** is malware that **attaches itself to a legitimate file or program** and typically requires some user/system action to execute and spread.

Key Points

- Requires a **host file/program**.
- Can modify, corrupt or delete data.
- Can replicate by infecting other files.

- Often spreads when an infected file is executed or shared.

Example

An infected executable is opened → the virus executes → other files may become infected.

Virus = Needs a host

2. Worm

A **worm** is malware that can **self-replicate and spread automatically**, often across networks, without needing to attach itself to another program.

Key Points

- **Self-replicating**
- Does not require a host file in the same way a virus does.
- Can spread rapidly across vulnerable systems.
- Often exploits network/software vulnerabilities.
- Can consume bandwidth and system resources.

Worm = Self-replicates + Spreads automatically

3. Trojan Horse

A **Trojan Horse** is malware that **disguises itself as legitimate or useful software** to trick the user into installing or executing it.

Key Points

- Relies heavily on **deception/social engineering**.
- Does **not self-replicate like a worm**.
- Once executed, it may steal data, install other malware or provide unauthorized access.
- Can appear as a legitimate application, attachment or download.

Example

A user downloads a program advertised as a useful utility, but the program secretly installs malicious software.

Trojan = Disguised as legitimate software

4. Ransomware

Ransomware is malware that **denies access to data or systems**, commonly by encrypting files, and demands a ransom or other action from the victim.

Key Points

- Commonly encrypts files.
- Makes data inaccessible to the victim.
- Attackers demand payment or another action.
- Can cause significant operational disruption.
- **Having reliable offline/isolated backups** is an important defensive measure.

Example

Files → Ransomware → Encrypted Files 

↓

Ransom Demand

Ransomware = Lock data/system + Demand ransom

Important

Paying a ransom **does not guarantee** recovery of files.

5. Spyware

Spyware is malware designed to **secretly monitor, collect or transmit information** about a user or system without proper authorization.

Key Points

- Can monitor user activity.
- May collect personal or sensitive information.
- Can track browsing behavior, credentials or other data depending on the spyware.
- Usually operates without the user's knowledge.

Spyware = Secretly monitors/collects information

6. Adware —

Adware is software that **displays unwanted advertisements**, often by monitoring or altering browsing behavior.

Key Points

- Commonly generates unwanted ads/pop-ups.
- Some adware can track user behavior.
- Not all adware is necessarily malware; classification depends on behavior, consent and context.

Adware = Unwanted advertising / tracking

7. Rootkit —

A **rootkit** is a collection of malicious software or techniques designed to **maintain privileged/unauthorized access while hiding its presence**.

Key Points

- Focuses on **stealth and persistence**.
- Can hide malicious processes, files or system activity.
- May operate at very low system levels.
- Can be difficult to detect and remove.

Rootkit = Hide + Maintain privileged access

8. Keylogger —

A **keylogger** is software or hardware designed to **record keystrokes** made by a user.

Key Points

- Can capture usernames, passwords, messages and other typed information.
- Often used for credential theft.
- Can exist as software or, in some cases, a physical device.

Keylogger = Records keystrokes

9. Botnet —

A **botnet** is a network of **compromised devices** controlled by an attacker or botnet operator.

Key Points

- Infected devices are often called **bots/zombies**.
- Devices can be remotely controlled as a group.
- Can be used for DDoS attacks, spam, credential attacks or other malicious activities.
- The device owner may be unaware that their device is part of the botnet.

Botnet = Network of compromised devices under attacker control

8.3 Network Attacks

Network attacks target **network communication, devices, protocols or sessions** with the goal of disrupting services, intercepting information, impersonating systems or gaining unauthorized access.

For revision, the most important thing is to understand **what is being targeted and what the attacker is trying to achieve**.

🌟 1. DoS — Denial of Service

A **DoS attack** attempts to make a system, server or network service **unavailable to legitimate users** by overwhelming it or exhausting its resources.

Key Points

- Usually originates from **one attacking system/source**.
- Consumes resources such as CPU, memory, bandwidth or connection capacity.
- Legitimate users may be unable to access the service.
- Primarily targets **Availability** in the CIA Triad.

Example

Attacker

↓↓↓↓↓↓↓

Web Server

×

Legitimate users cannot access service

DoS = Make service unavailable

🌐 2. DDoS — Distributed Denial of Service

A **DDoS attack** is a DoS attack in which traffic or requests come from **multiple distributed compromised or controlled systems**.

Key Points

- Uses many devices/sources simultaneously.
- Much harder to block than a simple single-source DoS.
- Can generate extremely large volumes of traffic or requests.
- Often uses a **botnet**.
- Primarily targets **Availability**.

3. MITM — Man-in-the-Middle

A **Man-in-the-Middle (MITM) attack** occurs when an attacker **positions themselves between two communicating parties** and intercepts, relays or potentially modifies their communication without their knowledge.

Key Points

- Attacker secretly gets between the communicating parties.
- Can potentially **intercept or modify data**.
- Can target authentication credentials or sensitive communication.
- Strong encryption and proper authentication help defend against MITM attacks.

Example

Alice ↔ Attacker ↔ Bob

↑

Intercepts/
potentially modifies
communication

MITM = Attacker gets in the middle of communication

4. Packet Sniffing

Packet sniffing is the process of **capturing and examining network packets** as they travel across a network.

Packet capture itself can be legitimate for troubleshooting and network administration, but attackers can misuse it to obtain sensitive information.

Key Points

- Captures network traffic.
- Can reveal information contained in **unencrypted traffic**.
- Particularly dangerous on insecure/shared networks.
- Encrypted protocols such as **HTTPS and SSH** make captured traffic much harder to interpret.

Example

An attacker captures packets on an insecure network and examines the traffic for sensitive information.

Packet Sniffing = Capture + Analyze packets

5. Eavesdropping

Eavesdropping is the unauthorized **listening to, monitoring or interception of communication**.

Key Points

- Focuses on secretly obtaining information from communication.
- Can occur through network traffic, wireless communication or other communication channels.
- Unencrypted communication is particularly vulnerable.
- Encryption helps protect the confidentiality of intercepted data.

Example

An attacker secretly monitors communication between two users on an unsecured network.

Eavesdropping = Secretly listen/monitor communication

Packet Sniffing vs Eavesdropping

Packet Sniffing → Technique of capturing/analyzing network packets

Eavesdropping → Unauthorized interception/monitoring of communication

They are closely related, and packet sniffing can be one technique used for eavesdropping.

6. Spoofing

Spoofing is the act of **falsifying or impersonating an identity, address or other identifying information** to make communication appear to come from a trusted or legitimate source.

Key Idea

Attacker pretends to be someone/something else.

Different types of spoofing target different identifiers.

7. IP Spoofing

IP Spoofing involves modifying the **source IP address** in network packets so that the packets appear to originate from another IP address.

Key Points

- Attacker falsifies the source IP address.
- Can be used in various attacks, including some DoS/DDoS techniques.
- Source-address authentication and network filtering can help mitigate certain forms of IP spoofing.
- A spoofed IP does **not necessarily mean the attacker has control of that IP address.**

Example

Actual Attacker IP → 192.168.1.50

Spoofed Source IP → 10.0.0.10

The packet claims:

“I came from 10.0.0.10.”

when it actually originated elsewhere.

IP Spoofing = Fake IP identity

8. MAC Spoofing

MAC Spoofing involves changing or impersonating the **MAC address** associated with a network interface.

Key Points

- Attacker changes the device's reported MAC address.
- Can be used to bypass poorly designed MAC-based access controls.
- Can facilitate impersonation or network attacks in certain environments.
- MAC addresses operate at the **Data Link Layer (Layer 2)**.

Example

Original MAC → AA:BB:CC:11:22:33

Spoofed MAC → AA:BB:CC:44:55:66

MAC Spoofing = Fake MAC identity

9. DNS Spoofing

DNS Spoofing involves providing **false DNS information** so that a domain name resolves to an incorrect or malicious destination.

Example

A user enters:

www.bank.com

but manipulated DNS information causes the request to resolve to:

Attacker-controlled server

The user may believe they are visiting the legitimate website.

Key Points

- Targets **DNS name resolution**.
- Can redirect users to malicious or fraudulent destinations.
- Can be associated with attacks such as **DNS cache poisoning**.
- DNS security mechanisms and HTTPS/TLS certificate validation can help reduce the impact.

DNS Spoofing = Fake DNS response → Wrong destination

10. Session Hijacking

Session Hijacking is an attack in which an attacker **takes control of an authenticated user's active session** by obtaining or exploiting session credentials/tokens.

Key Points

- Targets an already established authenticated session.
- Attacker may obtain a **session ID/token** and use it to impersonate the victim.
- Can result in unauthorized access without knowing the user's password.
- Secure session management, HTTPS, secure cookies and session expiration help reduce the risk.

Example

User → Login → Server

↓

Session Token

↓

Attacker

↓

Uses stolen token

↓

Impersonates User

Session Hijacking = Steal/abuse active session

8.4 Application & Web Attacks

Application and web attacks target **web applications, authentication mechanisms, user accounts, or application vulnerabilities**.

For Accenture-style questions, the key is to identify **what the attacker is exploiting and what the attack is trying to achieve**.

SQL Injection (SQLi)

SQL Injection is an attack in which an attacker inserts **malicious SQL input into an application's database query** to manipulate the query's behavior.

Key Points

- Targets applications that interact with databases.
- Usually occurs when **untrusted user input is improperly incorporated into SQL queries**.
- Can potentially allow unauthorized **reading, modification or deletion of database data**.
- Proper **parameterized queries/prepared statements** are a primary defense.

Simple Concept

User Input

↓

Web Application

↓

SQL Query

↓

Database

If input is handled unsafely, the attacker may influence the SQL query.

Example

A login form asks for:

Username + Password

If the application directly concatenates user input into a SQL query, specially crafted input may alter the intended query logic.

SQL Injection = Manipulate Database Query

 **Remember**

SQLi → Database

3. Brute Force Attack

A **Brute Force attack** attempts to gain unauthorized access by systematically trying **many possible passwords or credentials** until the correct one is found.

Key Points

- Does not necessarily rely on previously leaked passwords.
- May try combinations such as:
 - 123456
 - password
 - admin123
 - etc.
- Can be performed against login systems and other authentication mechanisms.
- Strong passwords, **MFA**, **rate limiting**, **account lockout** and **monitoring** help reduce the risk.

Example

Password attempt 1 ❌

Password attempt 2 ❌

Password attempt 3 ❌

...

Correct password ✅

Brute Force = Try many combinations

4. Credential Stuffing

Credential Stuffing is an attack in which attackers use **previously stolen username/password combinations** from one breach to attempt logins on other websites or services.

Why does it work?

Many users **reuse the same password across multiple websites**.

Example

Suppose an attacker obtains:

user@example.com : Password123

from a breached website.

The attacker then tries the same credentials on:

- Email
- Banking
- Social media
- Shopping websites

If the victim reused the password, another account may be compromised.

Prevention

- Use **unique passwords** for different accounts.
- Use a password manager.
- Enable **MFA**.
- Monitor for suspicious login activity.

Credential Stuffing = Reuse stolen credentials

5. Zero-Day Attack

A **Zero-Day vulnerability** is a security vulnerability that is **unknown to the vendor or for which no effective patch/fix was available at the time of exploitation**.

A **Zero-Day Attack/Exploit** refers to exploitation of such a vulnerability before an effective fix is available.

Why "Zero-Day"?

The vendor has effectively had:

0 days to provide a patch after becoming aware of the vulnerability/exploitation.

Key Points

- Exploits a previously unknown or unpatched vulnerability.
- Can be difficult to defend against because a patch may not yet exist.
- Organizations may use security monitoring, behavioral detection, network controls and other mitigations while a patch is unavailable.

Zero-Day = Exploitation before an effective fix is available

⚠ Important Distinction

Zero-day vulnerability → The vulnerability itself.

Zero-day exploit/attack → Using that vulnerability to attack a system.

🔒 6. Password Attacks

Password attacks are techniques used to **guess, steal, capture or crack passwords** to gain unauthorized access.

Common examples include:

◆ Brute Force

Tries many possible combinations.

◆ Dictionary Attack

Tries passwords from a **list of common words/passwords**.

◆ Credential Stuffing

Uses **stolen username/password pairs** from previous breaches.

◆ Password Spraying

Attempts a **small number of commonly used passwords across many accounts**, helping avoid triggering account lockouts that may occur with many attempts against one account.

◆ Keylogging

Uses a keylogger to capture what the victim types.

🛡️ Common Defenses

- Strong, unique passwords
- Password manager
- **MFA**
- Rate limiting
- Account lockout / throttling
- Monitoring and anomaly detection
- Secure password storage using appropriate password hashing/KDFs

📊 MASTER REVISION TABLE

Attack	What is Targeted?	Main Idea	Keyword
SQL Injection	Database / Web Application	Manipulate SQL queries through unsafe input	🔑 Database Query
XSS	Web Application / Browser	Execute malicious script in victim's browser	🌐 Script
Brute Force	Authentication	Try many possible passwords	🔑 Try All
Credential Stuffing	User Accounts	Reuse stolen credentials	🔑 Stolen Credentials
Zero-Day Attack	Vulnerable System/Application	Exploit before effective fix is available	🚫 No Fix Yet
Password Attacks	Authentication	Various techniques to obtain/crack passwords	🔑 Password

8.5 Social Engineering

Social Engineering is the use of **psychological manipulation, deception or human interaction** to trick people into revealing sensitive information, performing an action, granting access or bypassing security controls.

Social Engineering = Attack the human, not just the technology.

1. Social Engineering

Unlike technical attacks that directly exploit software or systems, social engineering exploits **human trust, emotions and behavior**.

Common psychological triggers

- **Urgency** → “Your account will be blocked today!”
- **Fear** → “Your account has been compromised.”
- **Authority** → “I’m calling from your bank/IT department.”
- **Curiosity** → “Look at this confidential document.”
- **Reward** → “You’ve won a prize!”
- **Trust** → Pretending to be a known person/organization.

Goal

Attackers may try to obtain:

- Passwords
- OTPs
- Banking information
- Personal information
- Access credentials
- Money
- Unauthorized system access

Social Engineering → Manipulate people to bypass security

2. Phishing

Phishing is a form of social engineering in which attackers use **fraudulent messages or communications** to trick victims into revealing sensitive information, clicking malicious links, opening malicious attachments or performing an unsafe action.

Common Channels

- Email
- Websites
- Messages
- Social media
- Other electronic communication

Example

You receive an email:

“Your bank account will be suspended. Verify your account immediately.”

The link leads to a fake login page designed to steal your credentials.

Common Warning Signs

- Urgent/threatening language
- Suspicious sender/address
- Unexpected attachments
- Suspicious links
- Requests for passwords/OTP
- Unusual payment requests

Phishing = Fraudulent message → Trick victim

3. Spear Phishing

Spear Phishing is a **targeted phishing attack** aimed at a specific individual or organization.

Unlike generic phishing, the attacker researches the victim and creates a **personalized message**.

Example

An employee receives an email appearing to come from their manager:

“Please urgently review this confidential salary document.”

The message uses the employee's real name, manager's name and organizational details.

Spear Phishing = Targeted + Personalized Phishing

Difference

Phishing → Broad/generic

Spear Phishing → Specific target

4. Whaling

Whaling is a form of spear phishing that specifically targets **high-profile individuals or senior executives**, such as CEOs, CFOs or senior managers.

Example

An attacker sends a fake email to a company's CFO pretending to be the CEO and requests an urgent financial transfer.

Whaling = Spear Phishing targeting senior/high-value individuals

Memory

Whale = Big target

5. Smishing

Smishing = SMS + Phishing

It is phishing performed through **SMS/text messages or similar messaging channels**.

Example

“Your parcel could not be delivered. Pay ₹50 to reschedule: [link]”

The link may lead to a fraudulent website designed to steal payment or personal information.

Common Signs

- Unexpected message
- Urgent request
- Suspicious link
- Request for payment or personal information

Smishing = Phishing through SMS/text

6. Vishing

Vishing = Voice + Phishing

It is phishing performed through **voice calls or phone-based communication**.

Example

Someone calls pretending to be from your bank:

“Your account has suspicious activity. Tell me the OTP to verify your identity.”

The attacker attempts to obtain sensitive information through the call.

Vishing = Phishing through Voice/Phone

Remember

**Bank/IT support asking for your OTP/password over an unexpected call
→ Major red flag**

7. Pharming

Pharming is an attack that **redirects users from a legitimate website/domain to a fraudulent or malicious destination**, often by

manipulating DNS-related mechanisms or compromising the user's/system's name-resolution process.

Unlike phishing, the victim **may not need to click a deceptive link**.

Example

You correctly enter:

www.example-bank.com

but DNS manipulation redirects you to a fake website.

Pharming = Redirect to fake destination

Phishing vs Pharming

Phishing → Tricks the user

Pharming → Manipulates/redirects the connection

8.6 Security Mechanisms

Security mechanisms are **tools, technologies and controls used to protect systems, networks, applications and data** from unauthorized access, malware, interception and other threats.

1. Firewall

A **firewall** is a security mechanism that **monitors and controls incoming and outgoing network traffic** according to predefined security rules.

Key Points

- Acts as a **barrier between trusted and untrusted networks**.
- Can allow or block traffic based on rules such as **IP address, port, protocol and connection state**.
- Helps prevent unauthorized network access.
- Can be implemented as hardware, software or a combination.

Example

Internet

↓

[FIREWALL]



Internal Network

If a firewall rule blocks incoming traffic on a particular port:

Traffic → Blocked

Firewall = Control network traffic

2. Antivirus

Antivirus software is designed to **detect, prevent and remove known or suspected malicious software**, particularly traditional computer viruses and other malware.

Key Points

- Scans files, programs and system activity.
- Can detect known malware using signatures and other detection techniques.
- May quarantine or remove detected threats.
- Modern antivirus products generally provide protection against **multiple types of malware**, not just viruses.

Antivirus = Detect + Prevent + Remove malware

3. Anti-Malware

Anti-malware software is a broader security solution designed to **detect, prevent and remove different types of malicious software**.

It may protect against:

- Viruses
- Worms
- Trojans
- Ransomware

- Spyware
- Other malware

Antivirus vs Anti-Malware

Today, the terms **antivirus** and **anti-malware** often overlap, because modern antivirus products commonly protect against many malware categories.

For exam-level distinction:

Antivirus → Traditionally focused on viruses

Anti-Malware → Broader protection against different malware

4. Encryption

Encryption is the process of transforming readable **plaintext** into **ciphertext** using a cryptographic algorithm and key, so that unauthorized parties cannot easily understand the data.

Main Purpose

Confidentiality

Used For

- Secure communication
- File encryption
- Database/storage protection
- VPN traffic
- HTTPS/TLS communication

Encryption = Protect data confidentiality

5. Access Control

Access Control is the process of **controlling who or what is allowed to access resources and what actions they are permitted to perform.**

Example

A college system may allow:

Student → View Marks

Faculty → View + Update Marks

Admin → Manage System

Each user gets only the permissions required for their role.

Common Access Control Models

- **DAC** → Discretionary Access Control
- **MAC** → Mandatory Access Control
- **RBAC** → Role-Based Access Control
- **ABAC** → Attribute-Based Access Control

 **Remember**

Access Control = Who can access what + what they can do

6. Authentication

Authentication is the process of **verifying the identity of a user, device or entity**.

Authentication = “Who are you?”

Common Methods

- Password
- OTP
- Biometrics
- Security token
- Multi-Factor Authentication (MFA)

Example

You enter your username and password.

The system verifies whether you are actually the claimed user.

Authentication → Identity verification

✅ 7. Authorization

Authorization determines **what an authenticated user or entity is allowed to access or perform.**

Authorization = “What are you allowed to do?”

Authentication	Authorization
Who are you?	What can you do?
Verifies identity	Determines permissions
Happens before authorization	Depends on established identity/permissions
Password, OTP, biometrics	Roles, permissions, access policies

🧠 Memory

Authentication = Identity

Authorization = Permission

🌐 8. VPN — Virtual Private Network

A **VPN** creates a **secure, encrypted connection/tunnel over an untrusted network**, such as the Internet.

Key Points

- Protects network traffic from unauthorized observation in appropriate VPN configurations.
 - Can securely connect remote users to private organizational networks.
 - Can connect geographically separated networks.
 - Often uses authentication and cryptographic protocols to protect the tunnel.
 - A VPN does **not make a user completely anonymous or automatically secure from every threat.**
-

9. Proxy Server

A **proxy server** acts as an **intermediary between a client and a destination server**.

Instead of the client communicating directly with the destination:

Client → Proxy → Destination

Uses

- Traffic filtering
- Access control
- Web caching
- Monitoring
- Content filtering
- Acting as an intermediary for client requests

The organization can use the proxy to apply web-access policies.

Proxy = Intermediary

9. Wi-Fi Security

9.1 Wi-Fi Basics

1. What is Wi-Fi?

Wi-Fi is a technology based on the **IEEE 802.11 family of standards** that allows devices to communicate wirelessly over radio waves, typically to form a local-area network.

It allows devices such as laptops, smartphones and IoT devices to connect to a network **without physical Ethernet cables**.

Wi-Fi = Wireless networking using IEEE 802.11

2. WLAN — Wireless Local Area Network

A **WLAN** is a **local-area network in which devices communicate wirelessly**, primarily using Wi-Fi/IEEE 802.11 technology.

Example

A college campus, office or home Wi-Fi network connecting:

Laptop + Smartphone + Printer + Smart TV

forms a **WLAN**.

Wi-Fi vs WLAN

WLAN = The wireless network

Wi-Fi = Technology used to provide wireless network connectivity

A WLAN can technically use wireless technologies other than Wi-Fi, so the terms are related but not perfectly identical.

3. Wireless Network

A **wireless network** is a network in which devices exchange data using **wireless communication technologies**, rather than requiring a physical cable for every connection.

Examples include:

- **Wi-Fi**
- Bluetooth
- Cellular networks
- Other wireless technologies

Wi-Fi is a type of wireless networking technology.

4. Wi-Fi Router

A **Wi-Fi router** is a network device that typically combines **routing, wireless access and other networking functions** to connect local devices to a network such as the Internet.

Main Functions

- Provides **Wi-Fi connectivity** to wireless devices.
- Routes traffic between the local network and other networks.

- Often provides **DHCP** to assign IP addresses.
 - Commonly performs **NAT** in home networks.
 - May include firewall and other security features.
-

5. Basic Wi-Fi Communication

Wi-Fi communication generally occurs between **wireless clients/stations** and a **wireless Access Point (AP)**.

Basic Flow

Laptop / Phone



Wi-Fi Radio



Access Point



Router / LAN



Internet

How it works — Basic Level

1. Device discovers the network

The wireless device scans for available Wi-Fi networks and their advertised information.

2. Device selects a network

The user chooses an SSID/network and provides the required authentication credentials when necessary.

3. Authentication & association

The device authenticates according to the configured Wi-Fi security method and associates with the AP.

4. Network configuration

The device typically receives network configuration such as an **IP address**, often through DHCP.

5. Data transmission

The device and AP exchange data wirelessly using Wi-Fi frames.

Important

Wi-Fi communication uses radio waves and Wi-Fi frames.

Don't confuse:

Wi-Fi → Wireless technology

Ethernet → Wired LAN technology

9.2 Wi-Fi Security Protocols

Wi-Fi security has evolved significantly:

WEP → WPA → WPA2 → WPA3

The important exam idea is:

WEP = Obsolete

WPA = Older transitional security

WPA2 = Major improvement, still widely deployed

WPA3 = Newer security generation

NIST guidance identifies WEP as inherently flawed and recommends moving toward newer Wi-Fi security; current NIST material also recommends WPA3 where possible.

1. WEP — Wired Equivalent Privacy

WEP was an early security mechanism designed to provide security for IEEE 802.11 wireless networks.

Key Points

- Uses the **RC4** stream cipher.
- Uses a relatively weak security design.
- Has serious cryptographic weaknesses.
- Can be cracked using practical attacks.
- **Should not be used for modern Wi-Fi security.**

2. WPA — Wi-Fi Protected Access

WPA was introduced as an improvement over WEP and served as a **transitional security solution** before the more robust WPA2 standard.

Key Points

- Improved security over WEP.
- Commonly associated with **TKIP** in its original implementation.
- Designed to address major weaknesses in WEP.
- It is now **legacy/obsolete for modern secure deployments**.

WPA = Better than WEP, but outdated today

9. WEP vs WPA vs WPA2 vs WPA3

Feature	WEP	WPA	WPA2	WPA3
Generation	Oldest	Transitional improvement	Major modern improvement	Newer generation
Security	✗ Very weak	⚠ Legacy	✓ Stronger	★ Strongest of these
Main Technology	RC4-based WEP	TKIP commonly associated	AES-CCMP	Modern WPA3 mechanisms
Personal Mode	Yes	Yes	WPA2-Personal	WPA3-Personal
Enterprise Mode	Limited/legacy	Yes	WPA2-Enterprise	WPA3-Enterprise
Current Recommendation	✗ Do not use	✗ Do not use	Use when WPA3 unavailable/required	★ Prefer where supported
Key Association	Obsolete	Legacy	Established	Modern

9.3 Wi-Fi Security Threats

Wi-Fi networks are particularly exposed to attacks because communication takes place over **radio signals**, which can potentially be observed by devices within transmission range. The major threats below focus on **interception, impersonation, unauthorized access and disruption**.

1. Eavesdropping

Eavesdropping is the unauthorized **listening to or monitoring of wireless communication** to obtain information being transmitted.

Key Points

- Attacker passively monitors wireless communication.

2. Packet Sniffing

Packet Sniffing is the process of **capturing and analyzing network packets** transmitted over a network.

In Wi-Fi environments, an attacker within wireless range may use a wireless sniffer to capture frames/traffic.

Key Points

- Focuses on **capturing network traffic**.

Packet Sniffing = Capture + Analyze traffic

3. Evil Twin Attack

An **Evil Twin** attack involves creating a **fake Wi-Fi access point that imitates a legitimate Wi-Fi network**, often using the same or a very similar SSID.

Example

Legitimate network:

College_WiFi

Attacker creates:

College_WiFi

A victim may accidentally connect to the attacker's AP.

Possible Risks

Once connected, the attacker may attempt to:

- Intercept traffic
- Collect sensitive information
- Redirect users to malicious websites
- Perform MITM-style attacks



4. Rogue Access Point

A **Rogue Access Point (Rogue AP)** is an **unauthorized wireless access point connected to or operating within an organization's network/environment.**

A rogue AP can:

- Provide unauthorized network access.
- Bypass organizational security controls.
- Capture or intercept traffic.
- Provide an insecure entry point into the network.

Rogue AP = Unauthorized Access Point

NIST notes that rogue APs can be deployed without IT approval and may be configured insecurely, creating significant security risks.

6. Wi-Fi Password Cracking

Wi-Fi Password Cracking refers to attempts to **discover or recover the password/passphrase used to protect a wireless network.**

The exact attack depends on the Wi-Fi security protocol and configuration.

Attackers may attempt to:

- Capture authentication-related information.
 - Guess likely passwords.
 - Perform offline password-guessing attacks where the protocol permits.
 - Exploit weak/obsolete security configurations.
-

7. Brute Force

A **Brute Force attack** systematically tries **many possible password combinations** until the correct one is found.

Example

Password 1 ❌

Password 2 ❌

Password 3 ❌

...

Correct Password ✅

In Wi-Fi Security

The attacker may attempt many possible passwords against the available authentication material, depending on the protocol and attack scenario.

Brute Force = Try many combinations

Important

Brute force is a **general password attack technique**, not a Wi-Fi-specific attack.

8. Dictionary Attack

A **Dictionary Attack** is a password-guessing technique that tries passwords from a **predefined list of likely/common passwords** rather than testing every possible combination.

Example

password

123456

welcome

qwerty

admin123

...

If the actual password is weak and appears in the attacker's wordlist, it may be discovered quickly.

Dictionary Attack = Try likely/common passwords

3. Attack → What Happens → Prevention

Attack	What Happens?	Key Prevention
Eavesdropping	Attacker listens to wireless communication	Use strong encryption / secure protocols
Packet Sniffing	Attacker captures network packets	Use encrypted protocols such as HTTPS/SSH
Evil Twin	Fake AP imitates a trusted Wi-Fi network	Verify SSID/network, use enterprise authentication where appropriate
Rogue AP	Unauthorized AP provides an unapproved network entry point	AP monitoring + network access controls
Deauthentication	Victim is forcibly disconnected from Wi-Fi	Protected Management Frames (PMF/802.11w) where supported
Wi-Fi Password Cracking	Attacker attempts to recover Wi-Fi credentials	Strong, unique password + modern WPA security
Brute Force	Many password combinations are attempted	Strong passwords + WPA3/appropriate authentication
Dictionary Attack	Common passwords/wordlists are tried	Strong, unpredictable password
MITM	Attacker gets between communicating parties	HTTPS/TLS + proper authentication + trusted networks

4. Secure Wi-Fi Best Practices

1. Use WPA3 Where Supported

Prefer:

WPA3-Personal / WPA3-Enterprise

For environments where WPA3 isn't available, appropriately configured WPA2 can still be used.

Avoid:

 **WEP**

 **Legacy WPA**

2. Use a Strong, Unique Wi-Fi Password

A good Wi-Fi password should be:

- Long
- Difficult to guess
- Unique
- Not based on personal information
- Not reused elsewhere

Never use simple passwords such as 12345678 or password.

3. Use Enterprise Authentication in Organizations

For colleges, companies and other larger environments:

WPA2-Enterprise / WPA3-Enterprise + 802.1X/EAP

is preferable to giving every employee one shared Wi-Fi password.

4. Secure the Router

Change:

- Default administrator username/password where applicable
 - Default Wi-Fi credentials
 - Unnecessary configuration settings
-

✖ 5. Use Guest Network / Segmentation

Keep untrusted devices separate from important systems.

Corporate Devices

|

Secure LAN

|

[Servers]

Guest Devices

|

Guest Wi-Fi

|

Internet

Guest Wi-Fi should not automatically provide access to internal resources.

🔒 6. Use Secure Protocols

Even on a secure Wi-Fi network, prefer:

HTTPS

SSH

TLS-protected services

Encrypted applications

This provides **defense in depth**.

7. Be Careful with Public Wi-Fi

Avoid connecting automatically to unknown networks.

Before connecting:

- Verify the network name when possible.
- Be suspicious of duplicate/similar SSIDs.
- Avoid sensitive transactions on suspicious networks.
- Prefer HTTPS.
- Use a trusted VPN when appropriate.

Public Wi-Fi ≠ Automatically Safe

9. Keep Devices Updated

Update:

- Router firmware
- Laptop/PC
- Smartphone
- Wi-Fi drivers
- Security software

Updates can patch vulnerabilities that attackers may exploit.

10. Don't Rely on MAC Filtering Alone

MAC filtering can be an additional control, but:

MAC addresses can be spoofed.

Therefore, MAC filtering should **not be treated as strong authentication**.

Extra But Important

What is VPN?

A **Virtual Private Network (VPN)** creates a **protected connection/tunnel over an untrusted network**, such as the Internet.

VPN = Protected tunnel over an untrusted network

How VPN Works

At a basic level:

User Device



VPN Client



Protected Tunnel



VPN Server / Network



Destination

The VPN uses **authentication and cryptographic mechanisms** to protect the tunnel according to the VPN technology being used.

Benefits of VPN

1. Secure Remote Access

Allows employees to securely access organizational resources remotely.

2. Protected Communication

Helps protect traffic from unauthorized observation while it travels through the VPN tunnel.

3. Secure Site-to-Site Connectivity

Can connect geographically separated networks securely over public infrastructure.

4. Privacy on Untrusted Networks

A VPN can reduce exposure of traffic to the local network when properly configured.

Limitations of VPN

1. Does Not Make You Completely Anonymous

A VPN does **not guarantee anonymity**.

2. VPN Provider Becomes Important

The VPN provider/network operator may be able to observe certain metadata or traffic characteristics depending on the VPN architecture and configuration.

3. Does Not Protect Against Every Attack

A VPN does not automatically protect you from:

- Phishing
- Malware
- Weak passwords
- Compromised websites
- Endpoint attacks

10.3 Firewall Concepts

What is a Firewall?

A **firewall** is a security mechanism that **monitors and controls network traffic according to predefined security rules**.

Firewall = Traffic Control

Hardware Firewall

A **hardware firewall** is a firewall implemented in a **dedicated network device/appliance**.

Example

Internet



[Firewall Appliance]



Internal Network

Common Use

- Organizations
- Data centers
- Enterprise networks
- Network perimeter protection

Software Firewall

A **software firewall** is implemented as software running on a computer or operating system.

It can control traffic associated with that particular host.

Example

A laptop's operating system firewall blocks unauthorized incoming connections.

Software Firewall = Host-level software protection

Network Firewall

A **network firewall** controls traffic **between networks or network security zones**.

Example:

Internet



Network Firewall



Corporate Network

Its rules can control traffic based on:

- IP addresses
- Ports
- Protocols
- Connection state
- Other traffic characteristics

Host-Based Firewall

A **host-based firewall** runs on an individual endpoint such as:

- Laptop
- Desktop
- Server

It controls traffic **to and from that specific host**.

Network vs Host-Based

Network Firewall → Protects/controls traffic between networks

Host-Based Firewall → Protects an individual system

10.4 Proxy & Network Security

Proxy Server

A **proxy server** acts as an **intermediary between a client and a destination server**.

Client



Proxy



Destination

Instead of directly communicating with the destination, the client sends the request through the proxy.

How Proxy Works

Step 1

Client sends a request.

Step 2

Proxy receives the request.

Step 3

Proxy applies its configured policies and may:

- Allow
- Block
- Filter
- Cache
- Modify/inspect, depending on type and configuration

Step 4

If permitted, the proxy forwards the request to the destination.

Client

↓

Proxy

↓

Internet / Server

↓

Proxy

↓

Client

Proxy = Intermediary



Proxy vs VPN

This comparison is **very important**.

Feature	Proxy	VPN
Basic Role	Intermediary for requests/traffic	Protected tunnel between endpoints/networks
Typical Scope	Often application/protocol-specific	Can protect traffic from a device/network depending on configuration
Encryption	✗ Not automatically provided	✓ Uses cryptographic protection for the VPN tunnel
Traffic Routing	Through proxy	Through VPN tunnel
Common Uses	Filtering, caching, access control	Secure remote access, protected connectivity
Security	Depends heavily on configuration/protocol	Protects the VPN tunnel, but not endpoints from every threat